# ELMFIRE Validation Guide


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## About This Guide

This guide compares ELMFIRE predictions with observations for the active validation cases. Each report describes the event, input data, modeling choices, comparison methods, and uncertainties. The opening summary presents the calculated decisions and the computational information available during report preparation. A completed simulation does not by itself establish validation.

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# 1 How to Read This Guide

Start with the validation summary for the recorded computational environment, declared parallel allocation, and scientific decision table. The case identifiers in the summary link to the corresponding case-report entries. The common landscape-validation evidence then explains the metrics and limitations shared by the current cases, followed by each complete standalone case report with its event-specific provenance, configuration, figures, results, and limitations.

Interpret PASS and FAIL only where a case defines a justified acceptance decision. CHARACTERIZED means that descriptive metrics exist without a justified pass threshold. Missing, unrun, incomplete, or otherwise non-decisional evidence is shown as NOT EVALUABLE. A completed narrative alone does not establish validation evidence. The environment recorded during report preparation is not necessarily the execution environment of every included case.

# 2 Validation summary

## 2.1 Computational environment recorded during report preparation

The following information describes the environment available during preparation of this guide. It does not establish the environment used for earlier simulations. An available compiler is not necessarily the compiler used to build the model. ELMFIRE was not executed to obtain these observations; an undeclared model release remains unknown.

| Configuration | Recorded value |
|---|---|
| Recorded at (Coordinated Universal Time) | 2026-10-21T20:25:11+00:00 |
| Declared ELMFIRE release | not declared |
| Operating system | macOS-27.0.1-arm64-arm-64bit-Mach-O |
| Processor architecture | arm64 |
| Message Passing Interface (MPI) implementation | mpirun (Open MPI) 5.0.9 |
| Available Fortran compiler | GNU Fortran (Homebrew GCC 15.2.0_1) 15.2.0 |
| Python version | CPython 3.14.6 |
| Scientific analysis libraries | NumPy 2.5.4; Matplotlib 3.11.2; PyYAML 6.0.3; Rasterio not installed; GDAL not installed; Pandas not installed; GeoPandas not installed; Shapely not installed |

## 2.2 Comparison type and declared parallel allocation

The comparison type distinguishes isolated and coupled processes or the spatial scale of validation. The process count is the allocation declared for each case, not a measurement of resources used in an earlier simulation. Missing declarations are not inferred.

| Case | Comparison type | MPI processes |
|---|---|---|
| caldor_2021 | Scientific comparison | 1 |
| camp_fire | Landscape-scale validation | 50 |
| czu_lightning_complex_2020 | Scientific comparison | 1 |
| glass_2020 | Scientific comparison | 1 |
| kincade_2019 | Scientific comparison | 1 |
| thomas_fire | Landscape-scale validation | 50 |
| tubbs_fire | Landscape-scale validation | 50 |

## 2.3 Scientific decision summary

The decisions below follow the established calculated metrics and case-specific acceptance criteria. Completing a simulation or preparing a report does not establish PASS. Missing or insufficient scientific evidence is reported as NOT EVALUABLE.

**Total cases:** 7 **PASS:** 0 **FAIL:** 0 **CHARACTERIZED:** 7 **NOT EVALUABLE:** 0

| Case | Category | Title | Decision |
|---|---|---|---|
| caldor_2021 | Scientific comparison | 2021 Caldor Fire reconstruction against GOFER hourly perimeters | **CHARACTERIZED** |

Continued on next page

| Case | Category | Title | Decision |
|---|---|---|---|
| camp_fire | Landscape-scale validation | 2018 Camp Fire landscape reconstruction | **CHARACTERIZED** |
| czu_lightning_complex_2020 | Scientific comparison | 2020 CZU Lightning Complex reconstruction against GOFER hourly perimeters | **CHARACTERIZED** |
| glass_2020 | Scientific comparison | 2020 Glass Fire reconstruction against GOFER hourly perimeters | **CHARACTERIZED** |
| kincade_2019 | Scientific comparison | 2019 Kincade Fire reconstruction against GOFER hourly perimeters | **CHARACTERIZED** |
| thomas_fire | Landscape-scale validation | 2017 Thomas Fire landscape reconstruction | **CHARACTERIZED** |
| tubbs_fire | Landscape-scale validation | 2017 Tubbs Fire landscape reconstruction | **CHARACTERIZED** |

**Decision vocabulary.** PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

## 3 Common landscape-validation evidence

The current Camp, Thomas, and Tubbs landscape cases were reconstructed from the supplied historical-fire datasets. Each is designed to compare a 50-member ELMFIRE ensemble with cumulative Visible Infrared Imaging Radiometer Suite (VIIRS) active-fire detections. The case reports retain the event-specific input provenance, namelist rationale, evidence status, and unresolved limitations.

Their common spatial metrics are Jaccard overlap, precision, recall, $F_1$, predicted-to-observed area ratio, and Brier score. Together these distinguish spatial mismatch, overgrowth, undergrowth, scale bias, and probabilistic error. Because the recovered archives define no acceptance thresholds and do not quantify VIIRS-hull uncertainty, completed runs are reported as CHARACTERIZED rather than PASS or FAIL unless justified thresholds are declared before inspecting results. Unresolved model-input provenance, VIIRS collection or granule provenance, and event-specific calibration history are stated explicitly in the standalone reports.

# 2021 Caldor Fire Reconstruction against GOFER Hourly Perimeters

## 1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Caldor Fire (2021) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 626 h, from the first GOES fire detection at 15 August 2021 at 01:00 UTC to the last GOFER perimeter at 10 September 2021 at 03:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. No model parameter was tuned against the GOFER observation of this fire; the spotting configuration was revised after comparisons for the Kincade and CZU Lightning Complex fires.

## 2 Reference event, experiment, and observations

GOFER records the Caldor Fire from 15 August 2021 at 01:00 UTC to 10 September 2021 at 03:00 UTC with 620 hourly perimeters, a final perimeter area of 994 km$^2$ and an official size of 221,835 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of $0.75 \pm 0.21$ km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 620 hourly perimeters of this fire on a 51.3-m grid in UTM zone 10N (EPSG:32610). The raster covers 995 km$^2$, consistent with the final GOFER perimeter area of 994 km$^2$; arrival times range from 7 to 626 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

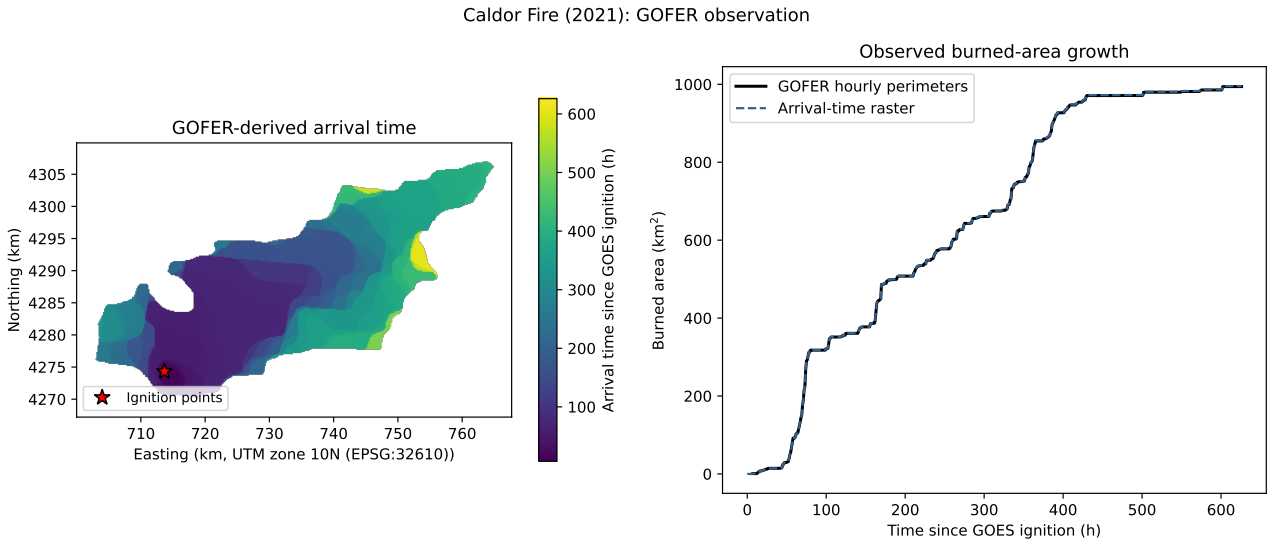



Figure 1: GOFER observation of the Caldor Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

## 3 Input data provenance and characterization

| Input | Source and version | Resolution, units and processing |
|---|---|---|
| Terrain | LANDFIRE elevation, slope and aspect [3] | 30 m; m and degrees; UTM zone 10N (EPSG:32610) |
| Surface fuels | LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.2.0 | 30 m categorical |
| Canopy | LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.2.0 | 30 m; percent, m, m and kg $m^{-3}$ |
| Weather | Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6] | 600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees |
| Fuel moisture | Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather | 600 m hourly; percent; derivation method not documented in the available case materials |
| Ignition | GOFER ignition points [2] | Points with hourly time stamps |
| Adjustment and initial state | Spread-rate adjustment factor and initial level set | Uniform value of one (no adjustment, nothing burned at the start) |

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 2,800 by 1,933 cells of 30 m (4,871 $km^2$) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 302 to 3,312 m (median 1,722 m) and slope has a median of 13° and a 95th percentile of 32°; the circular-mean aspect is 247°. By area the domain is 3.22% grass, 12.6% grass-shrub, 9.19% shrub, 49.2%

timber-understory, 14.4% timber litter, 0.0394% slash-blowdown and 11.4% non-burnable fuel models. Canopy cover has a median of 45% and canopy base height a median of 0.9 m.

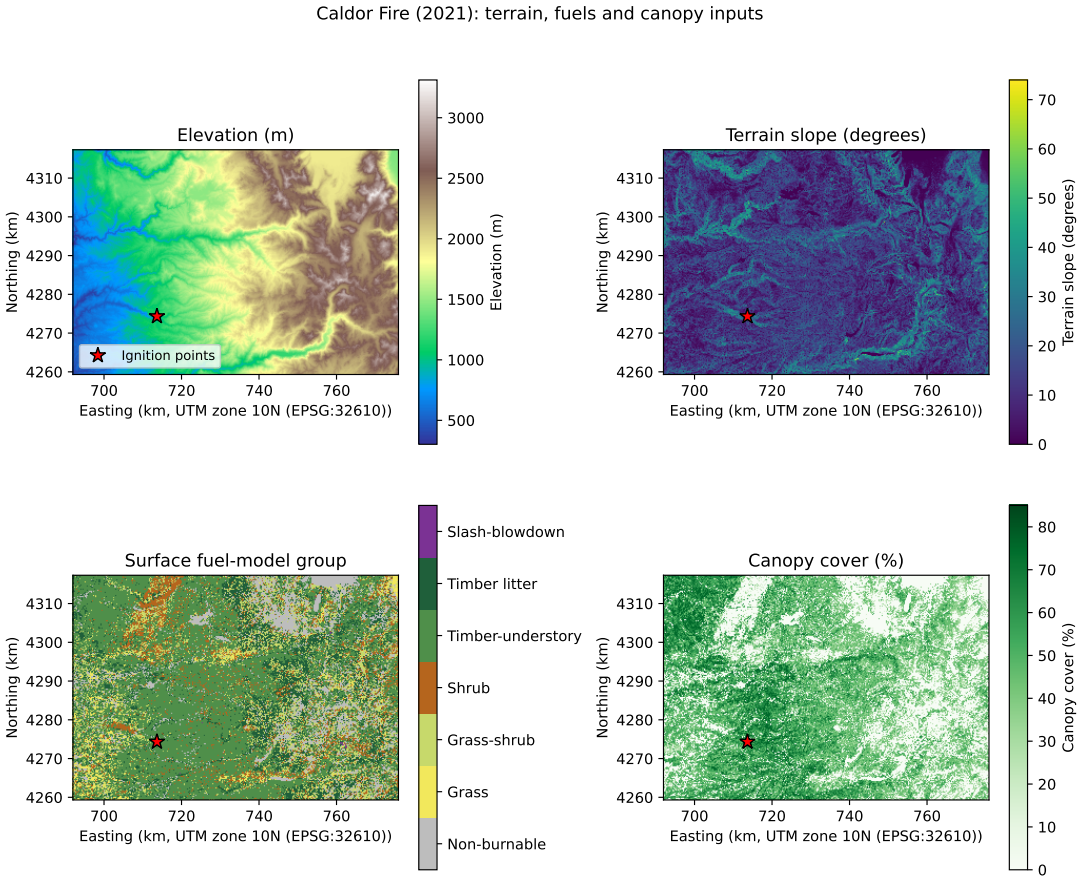



Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

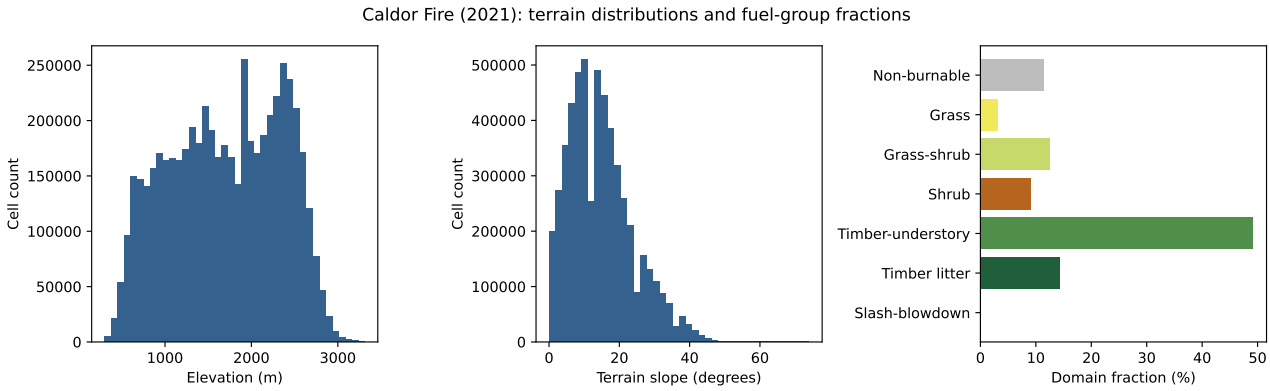



Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 621 hourly fields. Over all hours and cells the 20-ft wind speed has a median

of 3.89 mph and a 95th percentile of 11.4 mph; the circular-mean direction the wind blows from is 206° with a mean resultant length of 0.193 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 5.74% (5th percentile 3.2%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

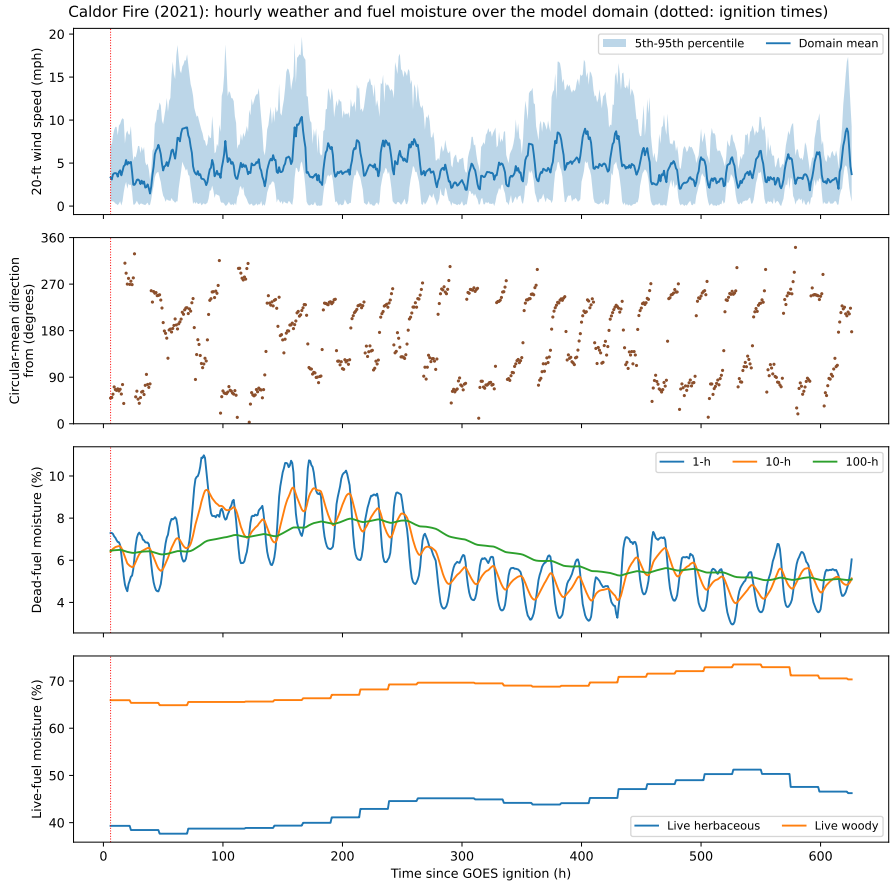



Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

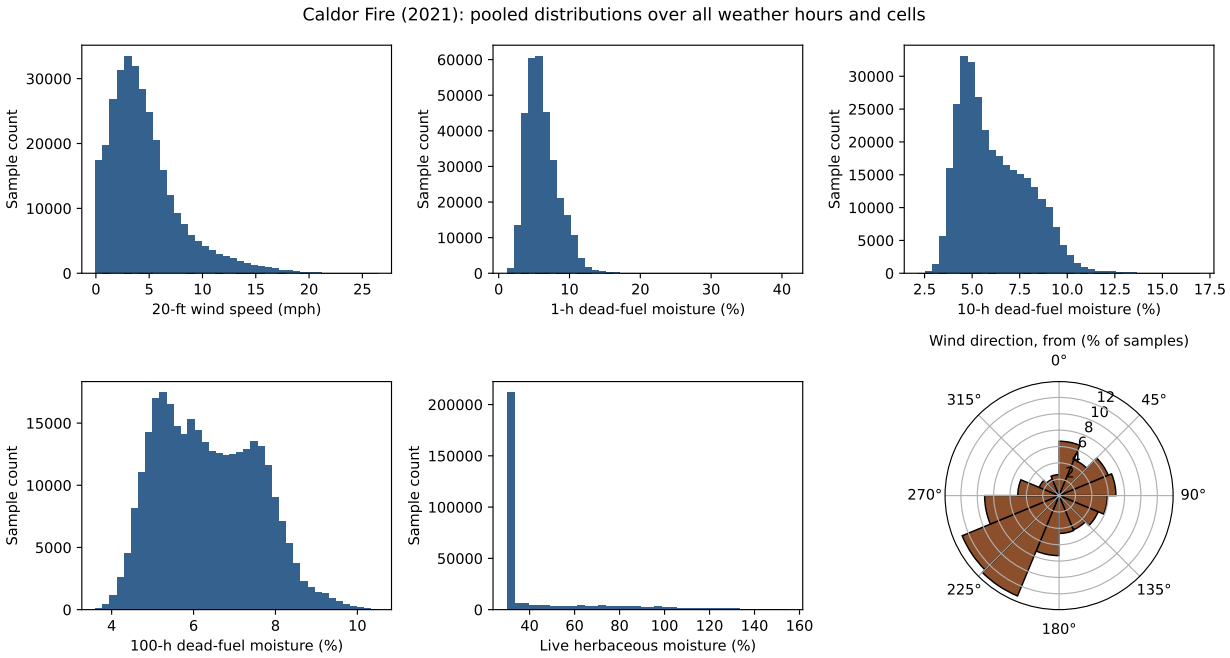



Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

## 4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 620 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft $\text{min}^{-1}$, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

| Choice | Basis | Evidence and limitation |
|---|---|---|
| Ignition points and times | documented | GOFER ignition record; no point required moving |
| Simulation start and stop | documented | First GOFER ignition point and last GOFER perimeter |
| Fuel vintage | derived | Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire |
| Weather and moisture | user-specified | RTMA analysis through CloudFire, as stated by the case owner |
| Crown fire | documented | ELMFIRE default crown-fire treatment, set explicitly |
| Firebrand spotting | user-specified | Enabled with ELMFIRE default spotting settings until calibration |
| Domain margin | user-specified | 10 km beyond the final GOFER perimeter |
| Single deterministic member | user-specified | No ensemble; stochastic spotting uses a fixed random seed |

Table 2: Basis of consequential configuration choices.

## 5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 6 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. The simulation results reported here were produced with ELMFIRE 1.1 at source revision `5ea7460d514b`, as recorded for the CZU Lightning Complex runs and assumed for the other results produced on the same system.

## 6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time $t$ the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap $J_t$ summarizes footprint agreement; precision $P_t$ diagnoses commission (simulated burning outside the observation), recall $R_t$ omission (observed burning the simulation misses), $F_t$ balances the two, and the area ratio $A_t$ exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

# 7 Simulation results and field-level comparison

The current status of this case is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Only results from earlier simulation configurations are available; the current case configuration has not been simulated.

## Single-ignition baseline without spotting

This result used a single point ignition at the first GOFER ignition point, firebrand spotting disabled and the ELMFIRE default crown-fire treatment, with the same inputs, stop time and numerical controls as the current configuration. The configuration file used was retained with this result.

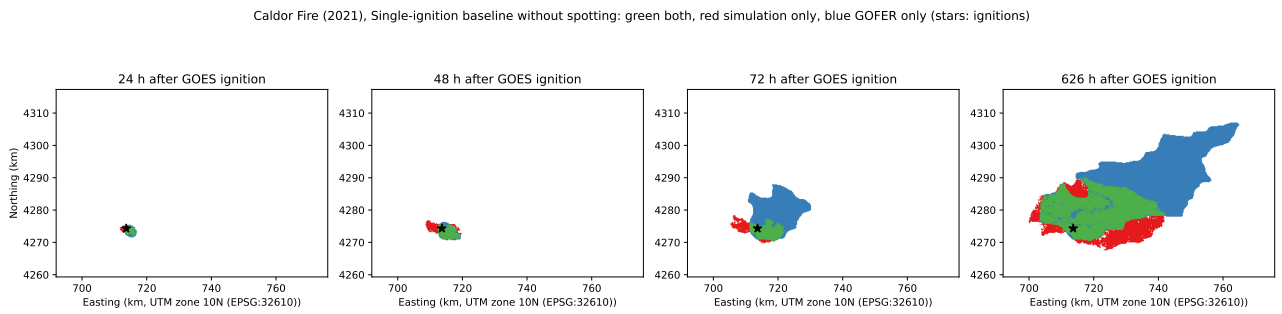



Figure 6: Single-ignition baseline without spotting: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

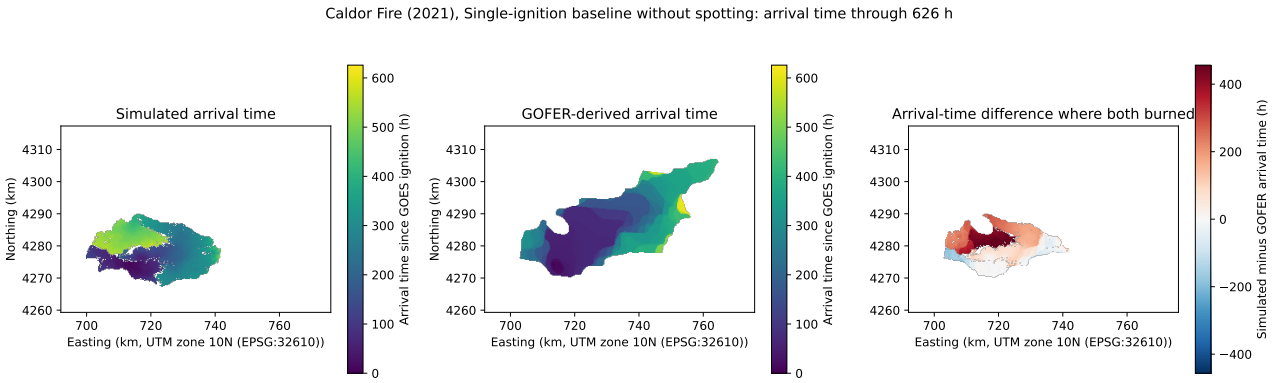



Figure 7: Single-ignition baseline without spotting: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

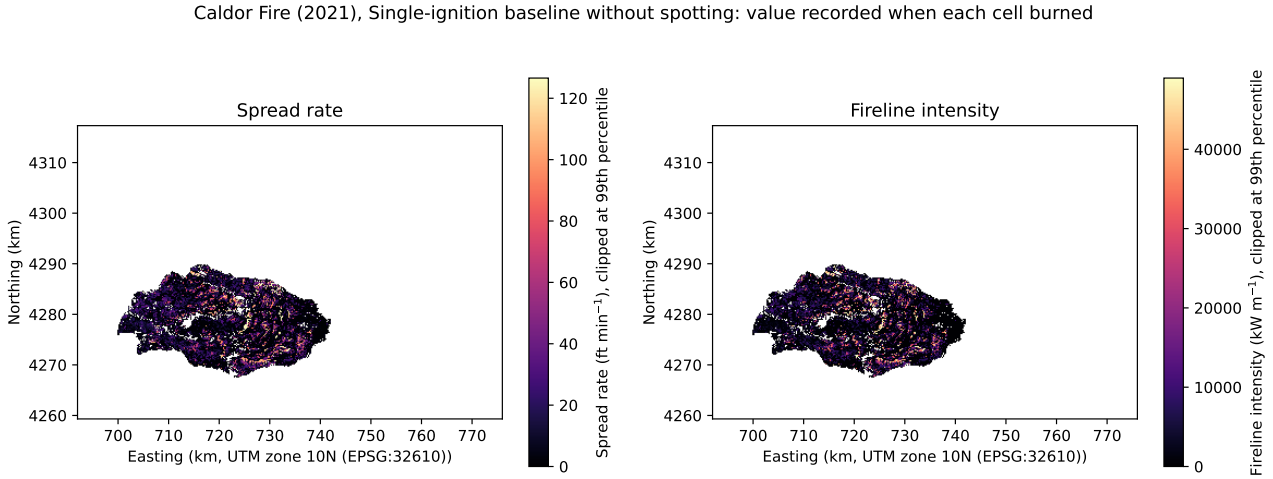



Figure 8: Single-ignition baseline without spotting: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

# 8 Integrated statistics and calculated metrics

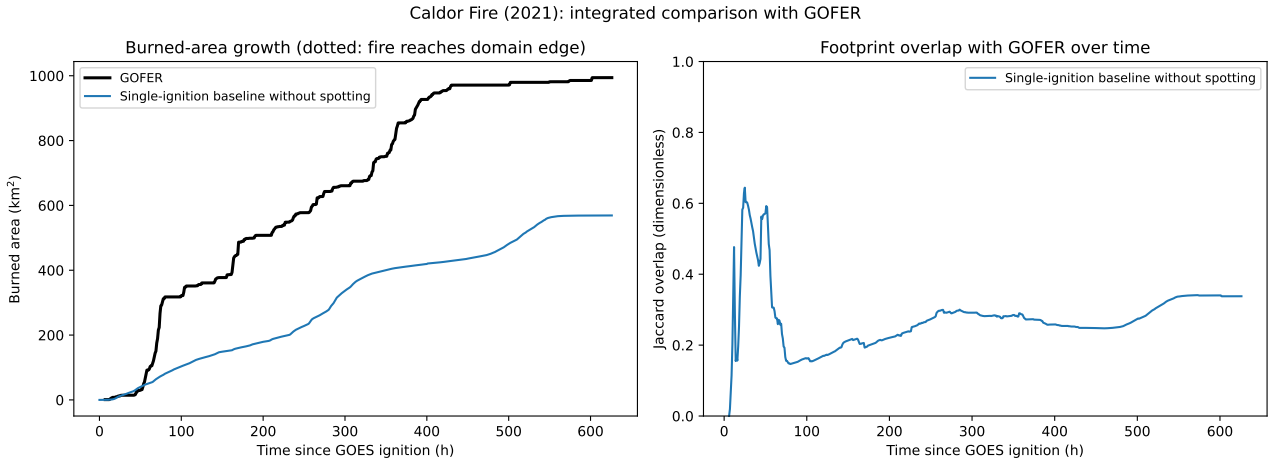



Figure 9: Observed (GOFER) and simulated burned-area growth, and the Jaccard overlap of each simulation with the GOFER footprint, evaluated hourly since the GOES ignition hour. Dotted lines mark the time each simulated fire first reaches the model-domain boundary.

## Single-ignition baseline without spotting

| Time (h) | GOFER (km$^2$) | Simulated (km$^2$) | Area ratio | Jaccard | Precision | Recall | $F_1$ |
|---|---|---|---|---|---|---|---|
| 24 | 12.1 | 11.1 | 0.918 | 0.63 | 0.81 | 0.74 | 0.77 |
| 48 | 29.2 | 37.5 | 1.29 | 0.57 | 0.64 | 0.83 | 0.73 |
| 72 | 216 | 68.7 | 0.318 | 0.2 | 0.68 | 0.22 | 0.33 |
| 626 | 995 | 569 | 0.572 | 0.34 | 0.69 | 0.4 | 0.51 |

Table 3: Footprint metrics of the single-ignition baseline without spotting against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

| Quantity | Value |
|---|---|
| Cells burned in both by 626 h | 438,997 |
| Mean arrival-time difference, simulated minus GOFER (h) | 150 |
| Median arrival-time difference (h) | 116 |
| Mean absolute arrival-time difference (h) | 172 |
| Root-mean-square arrival-time difference (h) | 227 |
| Simulated end since GOES ignition (h) | 626 |
| Propagation stalled before the planned end | no |
| First contact with domain boundary (h) | none |
| Final simulated area (ac) | 140,605 |
| Final crown-fire area (ac) | 69,820 |

Table 4: Arrival-time and integrated statistics of the single-ignition baseline without spotting. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Precision exceeds recall at the final comparison time, so the simulated footprint lies largely inside the observation but misses much of it: the simulation undergrows or follows a displaced trajectory. The simulated burned area is 0.57 times the observed area at 626 h. Where both burned, the simulated fire arrives on average 149.5 h later than the observed fire (mean absolute difference 172.3 h).

## 9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

## 10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Caldor Fire. Inputs and the observation are characterized in full; the current scientific status is **CHARACTERIZED** with 1 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

## References


[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

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[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

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# 2018 Camp Fire Landscape Reconstruction

## 1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2018 Camp Fire Landscape Reconstruction through 36 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

## 2 Reference event, experiment, and observations

The configured event start, expressed in Pacific standard time (PST), is 8 November 2018 at 14:00 PST. The comparison instant is 10 November 2018 at 02:00 PST (10:00 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 2260 point records in the geographic coordinate system defined by EPSG:4326; 2017 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

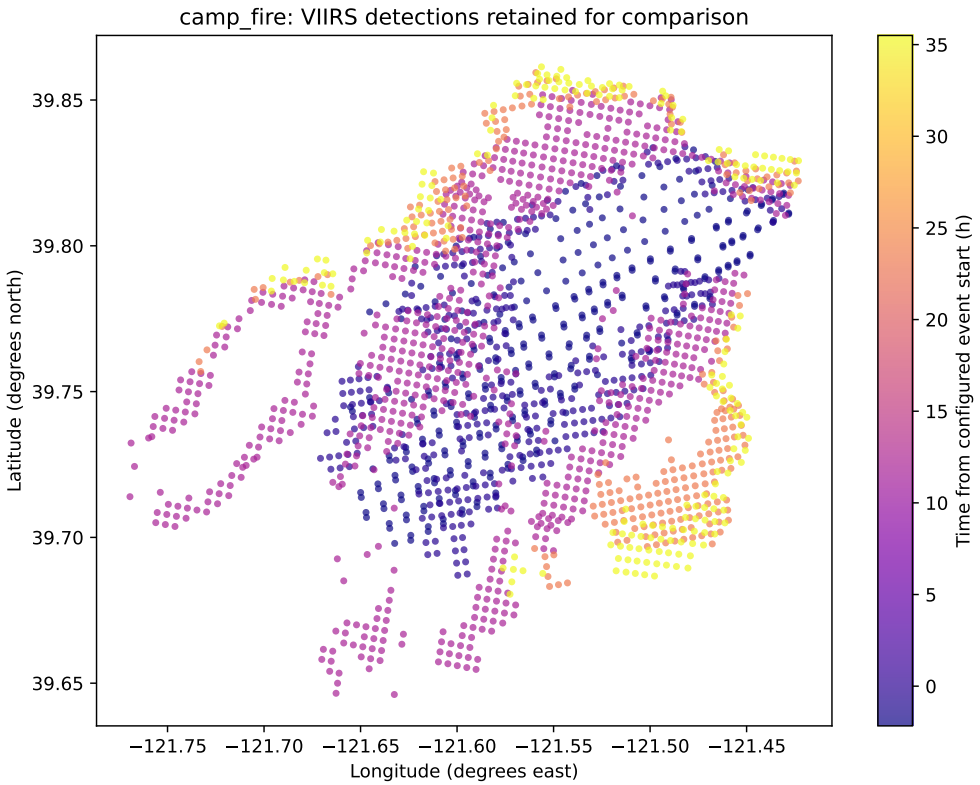



Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

## 3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `0b0e8f7ce3bbea4665269ff7f5b88dd373f3d9210b3fa4ddc7916b610f724033`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

| Field | Minimum | Mean | Standard deviation | Maximum |
|---|---|---|---|---|
| Elevation | 30 | 470.9 | 365.7 | 1620 |
| Slope | 0 | 11.74 | 10.32 | 63 |
| Wind speed | 0 | 8.313 | 6.306 | 33.35 |
| 1-h moisture | 6 | 6 | 0 | 6 |
| 10-h moisture | 7 | 7 | 0 | 7 |
| 100-h moisture | 8 | 8 | 0 | 8 |

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

No additional active-range anomaly was identified by the structural and statistical checks; upstream plausibility remains unverified because product metadata are absent.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 329.6° and 2.253°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most $512 \times 512$ cells per band, and are labeled accordingly.

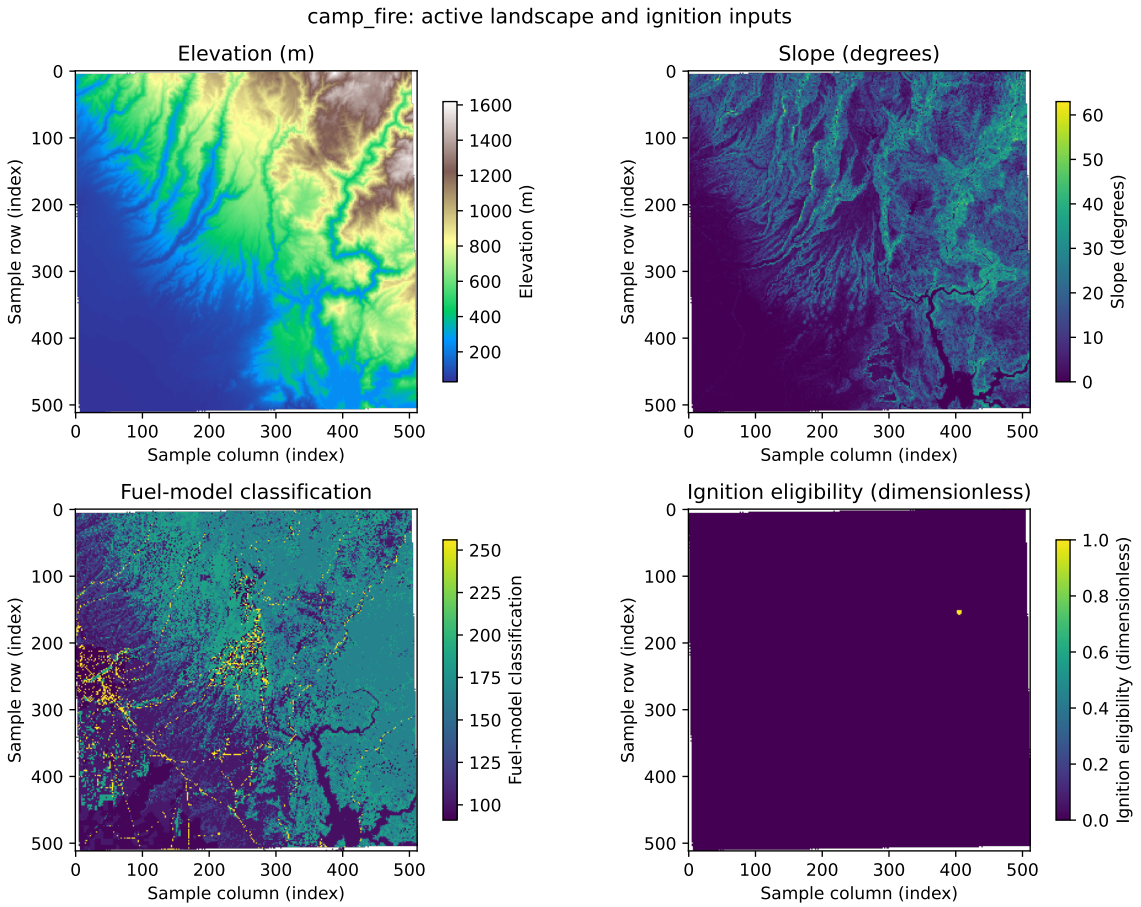



Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

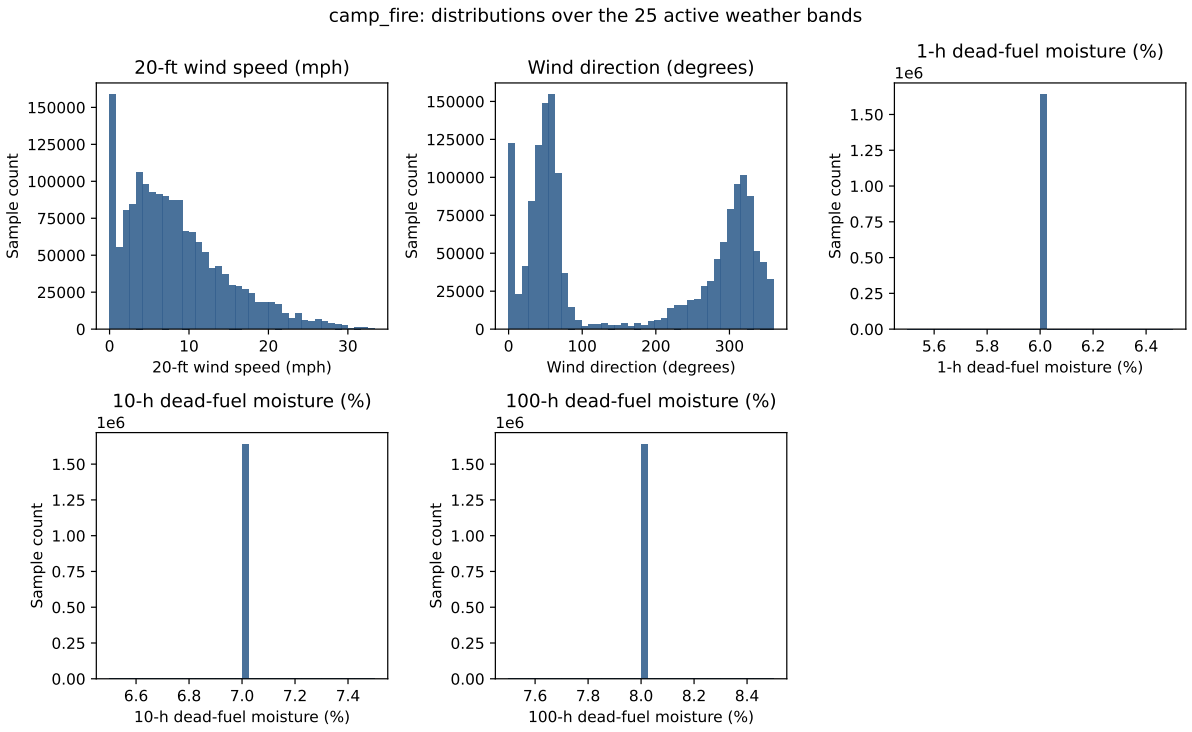



Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

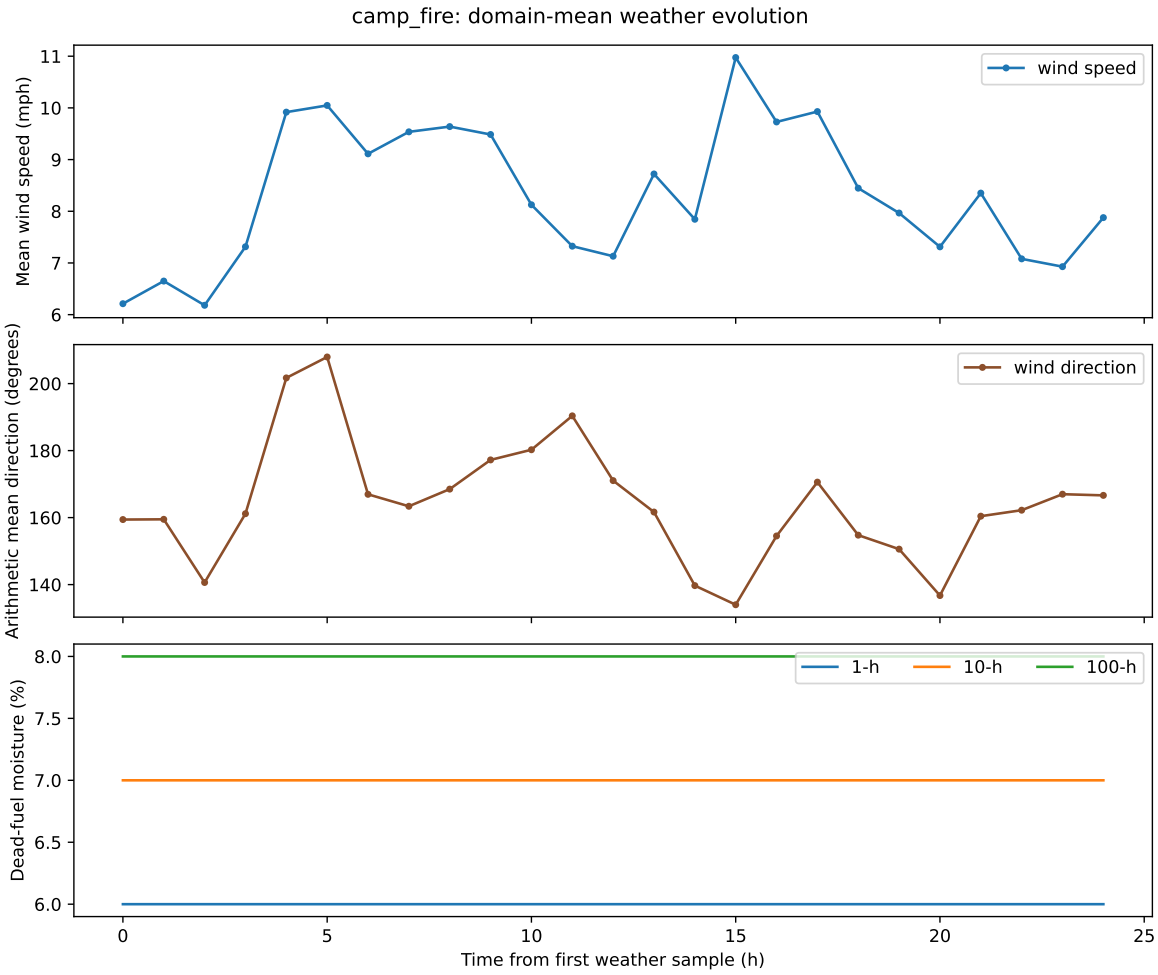



Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

## 4 Model configuration and rationale

The active domain is 1457 by 1476 cells at 30 m in EPSG:26910. Weather is 37-band weather rasters; bands 1–25 are selected at one-hour intervals, and the ignition treatment is random ignition sampling from the archived ignition mask. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

| Choice | Evidence class | Basis and limitation |
|---|---|---|
| Landscape, weather, ignition, Monte Carlo, spotting, and WUI values | documented | Retained from the received archive; upstream derivation and event-specific tuning history are unknown. |
| Model-setting compatibility | derived | Required by the choices supported by the inspected model formulation; no physical values were tuned. |
| Comparison time | documented | Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent. |
| $p \geq 0.50$ footprint | selected for this comparison | Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use. |
| VIIRS hull ratio 0.2 | selected for this comparison | Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter. |

Table 2: Provenance of consequential configuration and comparison choices.

## 5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

## 6 Validation metrics and comparison method

For member $i$, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and $t_c$ the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c] \,, \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support $O$, intersection $I = |P \cap O|$, and valid-cell count $M$, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and $F_1$ balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

## 7 Simulation results and field-level comparison

The current scientific status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

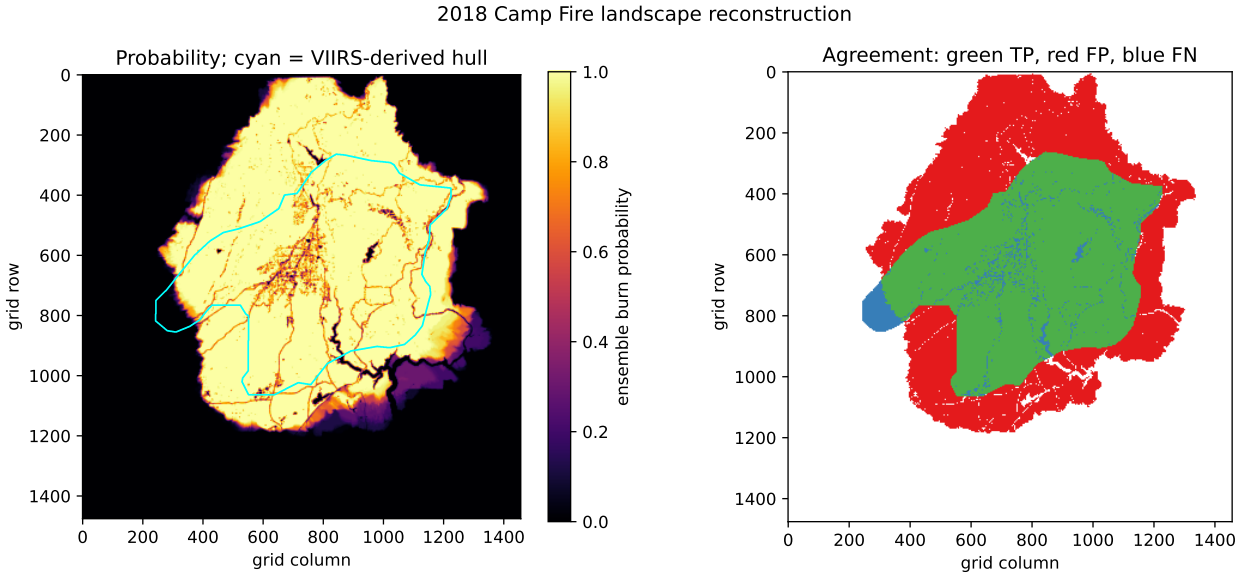



Figure 5: Ensemble burn probability, VIIRS-derived support, and cell-level true-positive, false-positive, and false-negative regions.

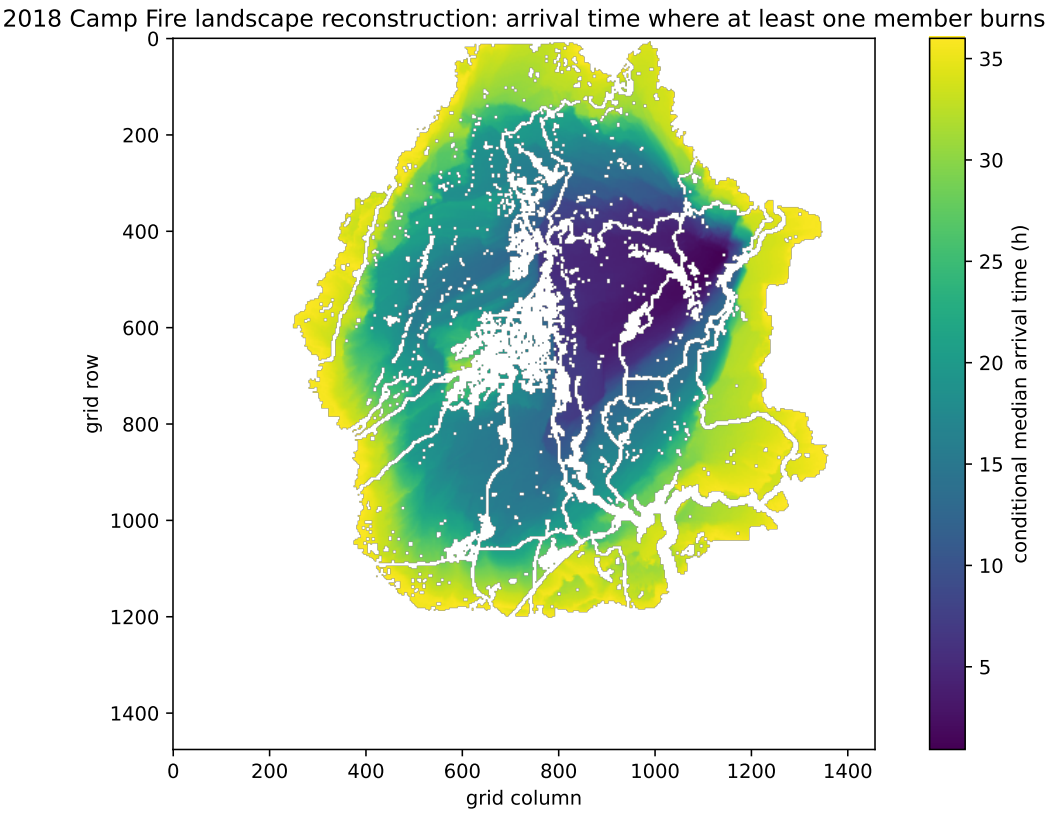



Figure 6: Conditional median arrival time among members that burn each cell by the comparison time. Conditioning is stated because it omits no-burn members.

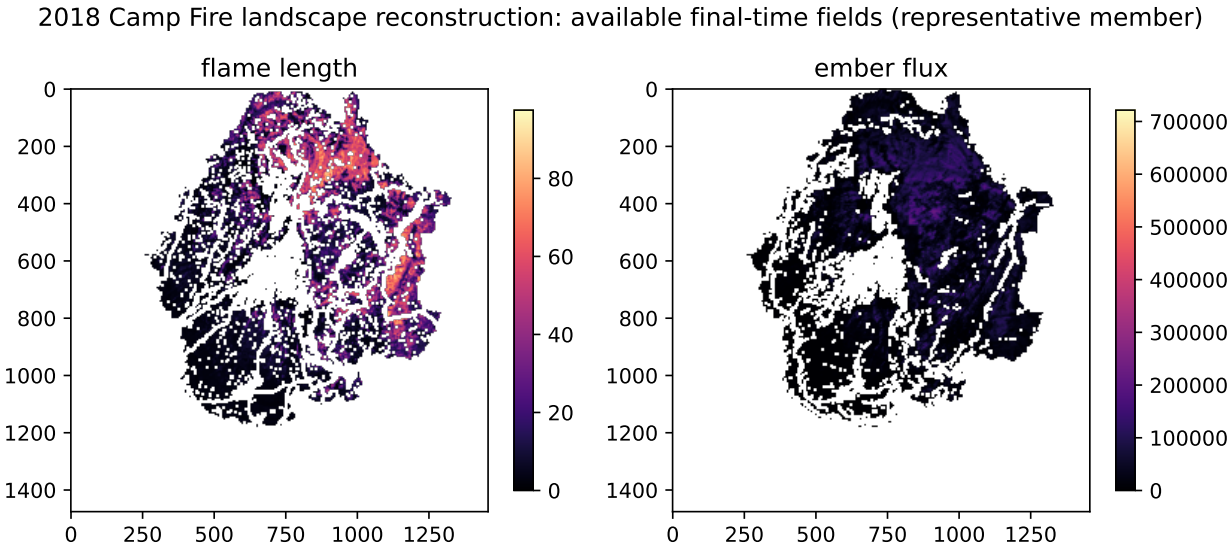



Figure 7: Applicable final-time two-dimensional fields found by the postprocessor. Omission of this panel means no recognized optional field was available; it does not imply a zero field.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported

and must not be inferred from WUI input rasters alone.

## 8 Integrated statistics and calculated metrics

| Metric | Current value |
|---|---|
| Jaccard overlap | 0.4858 |
| Precision | 0.5061 |
| Recall | 0.9236 |
| $F_1$ score | 0.6539 |
| Brier score | 0.197 |
| Predicted/observed area ratio | 1.825 |
| Predicted area (km$^2$) | 720.5 |
| Observation-support area (km$^2$) | 394.8 |
| Ensemble members found | 50 |

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

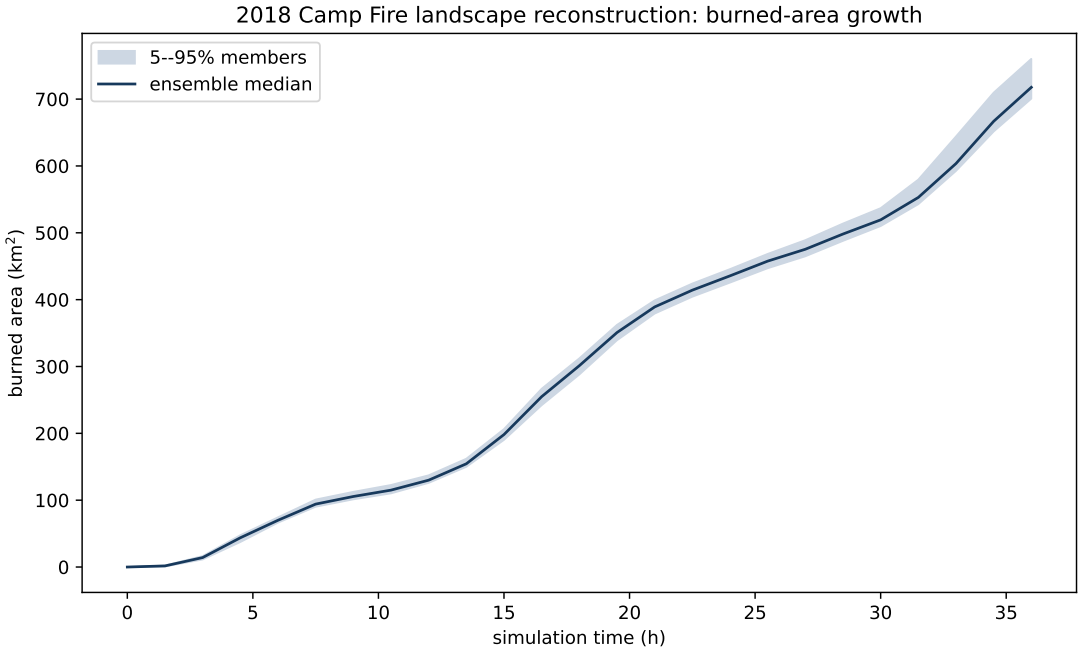



Figure 8: Median and 5th–95th percentile ensemble burned-area growth.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

# 9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

# 10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **CHARACTERIZED**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

# 2020 CZU Lightning Complex Reconstruction against GOFER Hourly Perimeters

## 1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the CZU Lightning Complex (2020) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 134 h, from the first GOES fire detection at 16 August 2020 at 15:00 UTC to the last GOFER perimeter at 22 August 2020 at 05:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. The firebrand-spotting configuration was changed after the case owner inspected comparisons of this fire with GOFER, so results for this fire are not independent of that choice.

## 2 Reference event, experiment, and observations

GOFER records the CZU Lightning Complex from 16 August 2020 at 15:00 UTC to 22 August 2020 at 05:00 UTC with 125 hourly perimeters, a final perimeter area of 283 km$^2$ and an official size of 86,509 ac, and 3 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of $0.75 \pm 0.21$ km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 125 hourly perimeters of this fire on a 50-m grid in UTM zone 10N (EPSG:32610). The raster covers 283 km$^2$, consistent with the final GOFER perimeter area of 283 km$^2$; arrival times range from 10 to 134 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

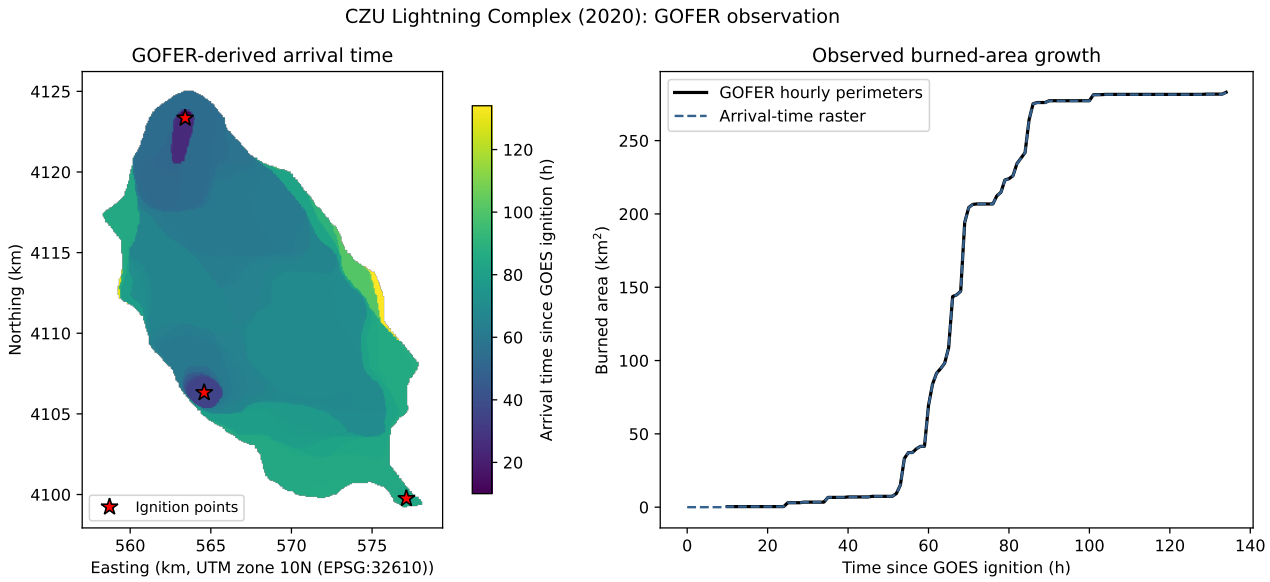



Figure 1: GOFER observation of the CZU Lightning Complex: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

## 3 Input data provenance and characterization

| Input | Source and version | Resolution, units and processing |
|---|---|---|
| Terrain | LANDFIRE elevation, slope and aspect [3] | 30 m; m and degrees; UTM zone 10N (EPSG:32610) |
| Surface fuels | LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.1.0 | 30 m categorical |
| Canopy | LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.1.0 | 30 m; percent, m, m and kg m$^{-3}$ |
| Weather | Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6] | 600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees |
| Fuel moisture | Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather | 600 m hourly; percent; derivation method not documented in the available case materials |
| Ignition | GOFER ignition points [2] | Points with hourly time stamps |
| Adjustment and initial state | Spread-rate adjustment factor and initial level set | Uniform value of one (no adjustment, nothing burned at the start) |

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 1,333 by 1,533 cells of 30 m (1,839 km$^2$) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from -1 to 985 m (median 148 m) and slope has a median of 9° and a 95th percentile of 31°; the circular-mean aspect

is 223°. By area the domain is 5.31% grass, 7.08% grass-shrub, 1.54% shrub, 9.06% timber-understory, 37.6% timber litter, 0.482% slash-blowdown and 38.9% non-burnable fuel models. Canopy cover has a median of 0% and canopy base height a median of 0 m. Offshore, 283,496 cells outside LANDFIRE coverage were assigned LANDFIRE open-water values; they are non-burnable.

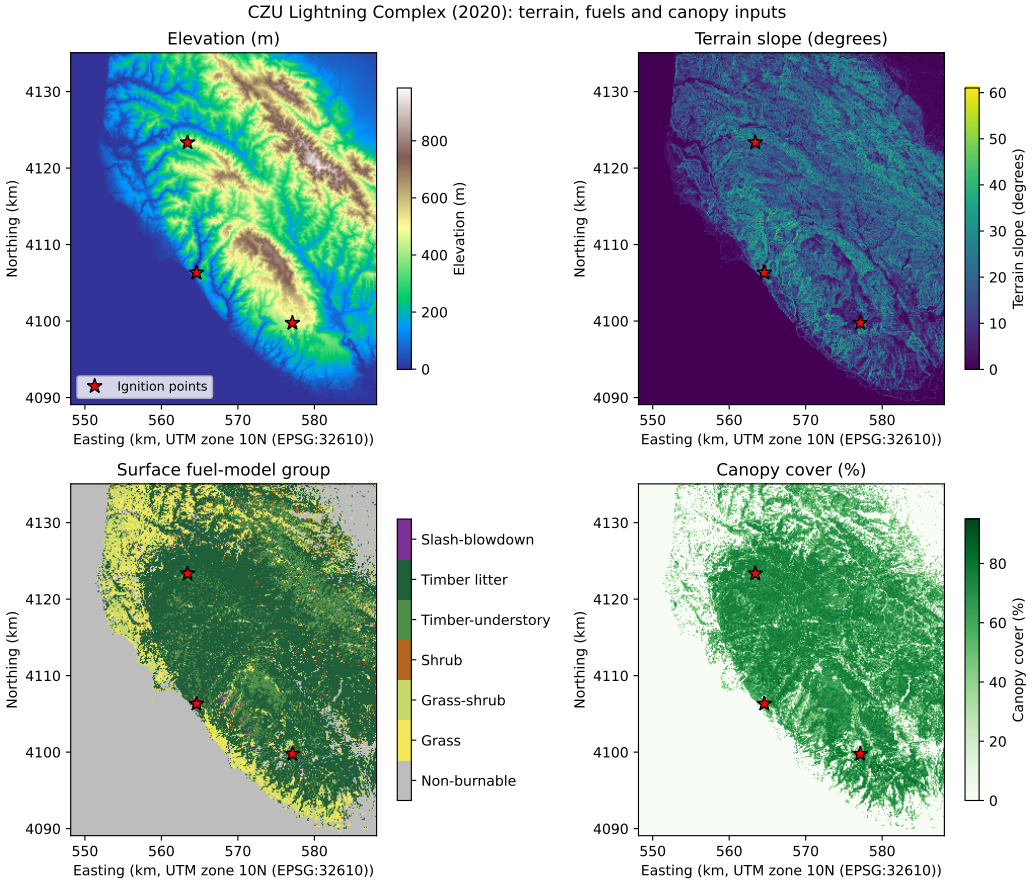



Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

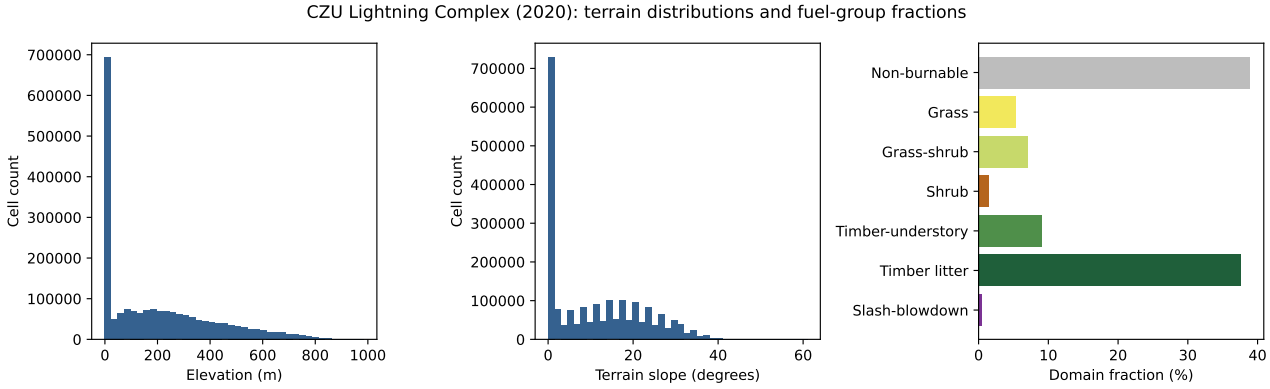



Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 126 hourly fields. Over all hours and cells the 20-ft wind speed has a median of 5.44 mph and a 95th percentile of 16.1 mph; the circular-mean direction the wind blows from is 328° with a mean resultant length of 0.732 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 15.3% (5th percentile 5.01%), and live herbaceous and woody moisture have medians of 160% and 143%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

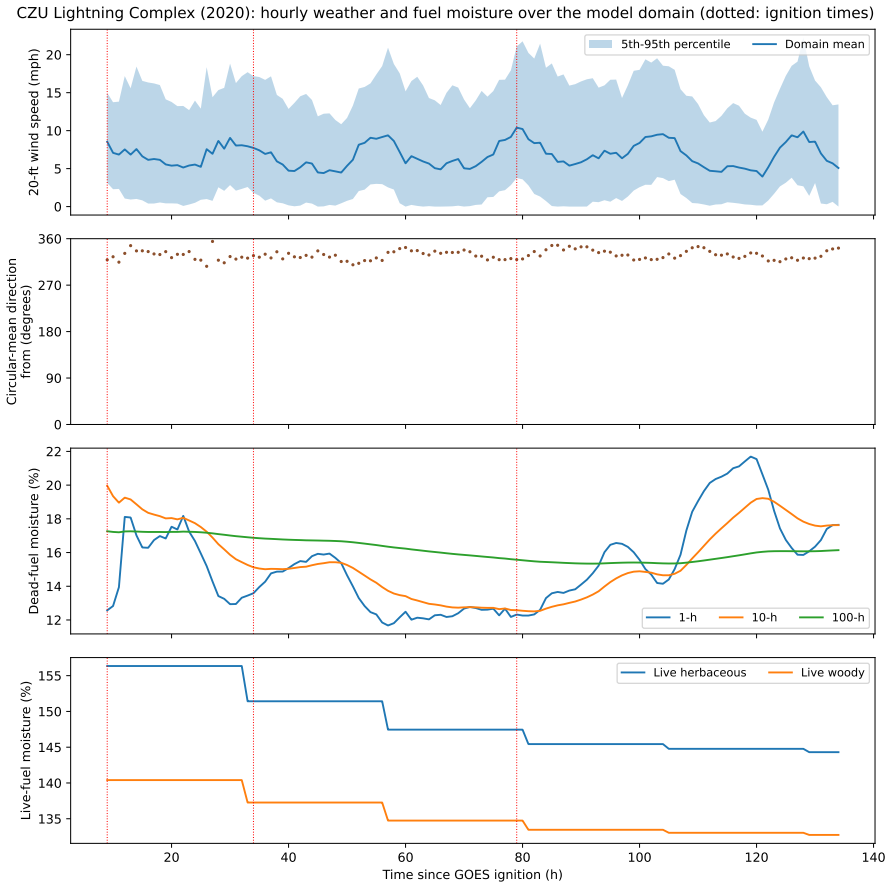



Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

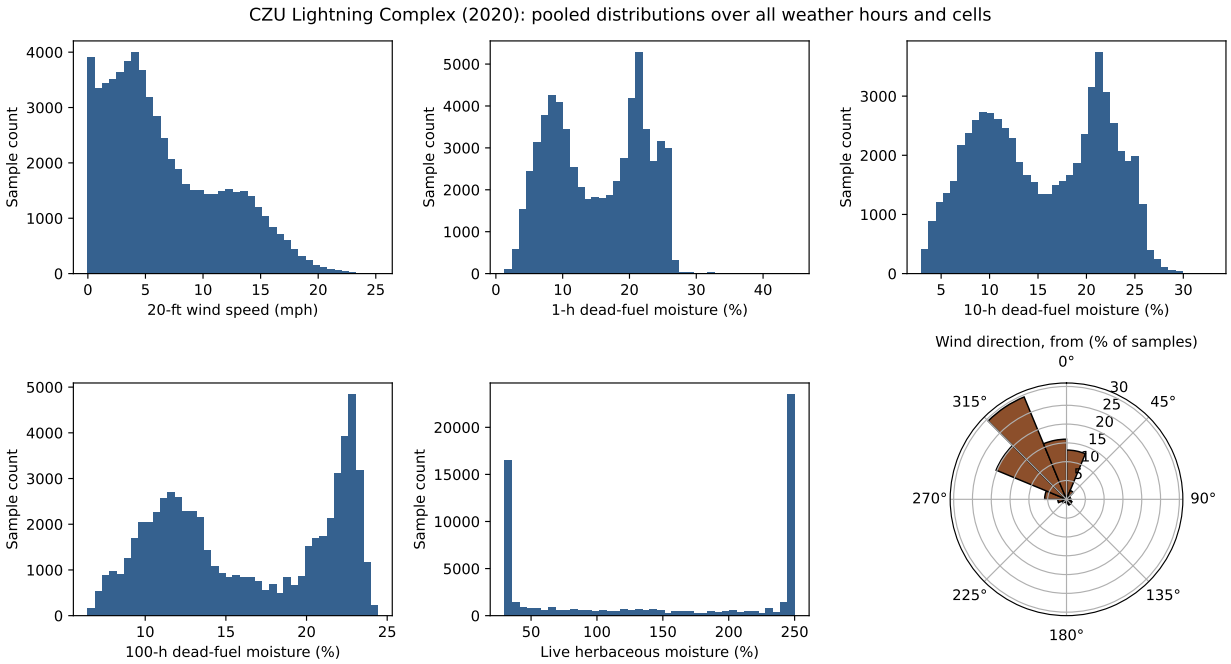



Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

## 4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 125 h, to the last GOFER perimeter, with 3 point ignitions, each applied at its observed GOFER time. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft $\mathrm{min}^{-1}$, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

| Choice | Basis | Evidence and limitation |
|---|---|---|
| Ignition points and times | documented | GOFER ignition record; no point required moving |
| Simulation start and stop | documented | First GOFER ignition point and last GOFER perimeter |
| Fuel vintage | derived | Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire |
| Weather and moisture | user-specified | RTMA analysis through CloudFire, as stated by the case owner |
| Crown fire | documented | ELMFIRE default crown-fire treatment, set explicitly |
| Firebrand spotting | user-specified | Enabled with ELMFIRE default spotting settings until calibration |
| Domain margin | user-specified | 10 km beyond the final GOFER perimeter |
| Single deterministic member | user-specified | No ensemble; stochastic spotting uses a fixed random seed |

Table 2: Basis of consequential configuration choices.

## 5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 9 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. The simulation results reported here were produced with ELMFIRE 1.1 at source revision `5ea7460d514b`, as recorded for the CZU Lightning Complex runs and assumed for the other results produced on the same system.

## 6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time $t$ the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap $J_t$ summarizes footprint agreement; precision $P_t$ diagnoses commission (simulated burning outside the observation), recall $R_t$ omission (observed burning the simulation misses), $F_t$ balances the two, and the area ratio $A_t$ exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

# 7 Simulation results and field-level comparison

The current status of this case is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Only results from earlier simulation configurations are available; the current case configuration has not been simulated.

## Single-ignition baseline without spotting

This result used a single point ignition at the first GOFER ignition point, firebrand spotting disabled and the ELMFIRE default crown-fire treatment, with the same inputs, stop time and numerical controls as the current configuration. The configuration file was not retained with this result; the configuration is documented from the case owner's preparation record. This result was produced before offshore cells outside LANDFIRE coverage were filled with open-water values; those cells then held a fill value read as data. They lie offshore beyond a band of open water that the simulated fire did not reach, but the result is not from the corrected inputs.

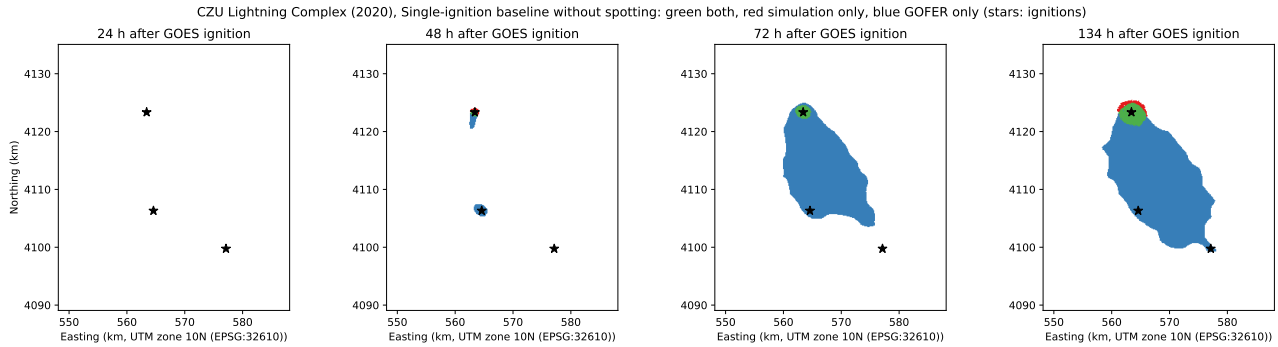



Figure 6: Single-ignition baseline without spotting: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

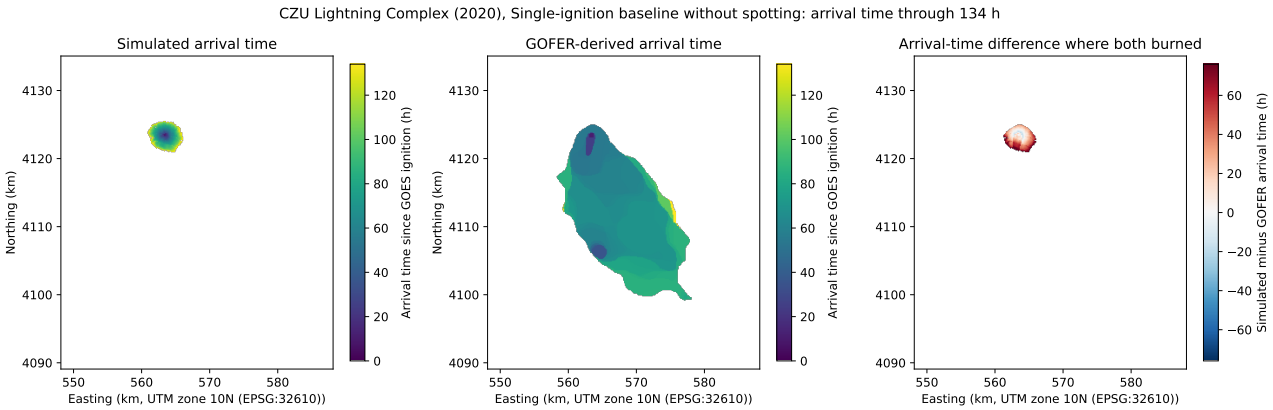



Figure 7: Single-ignition baseline without spotting: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

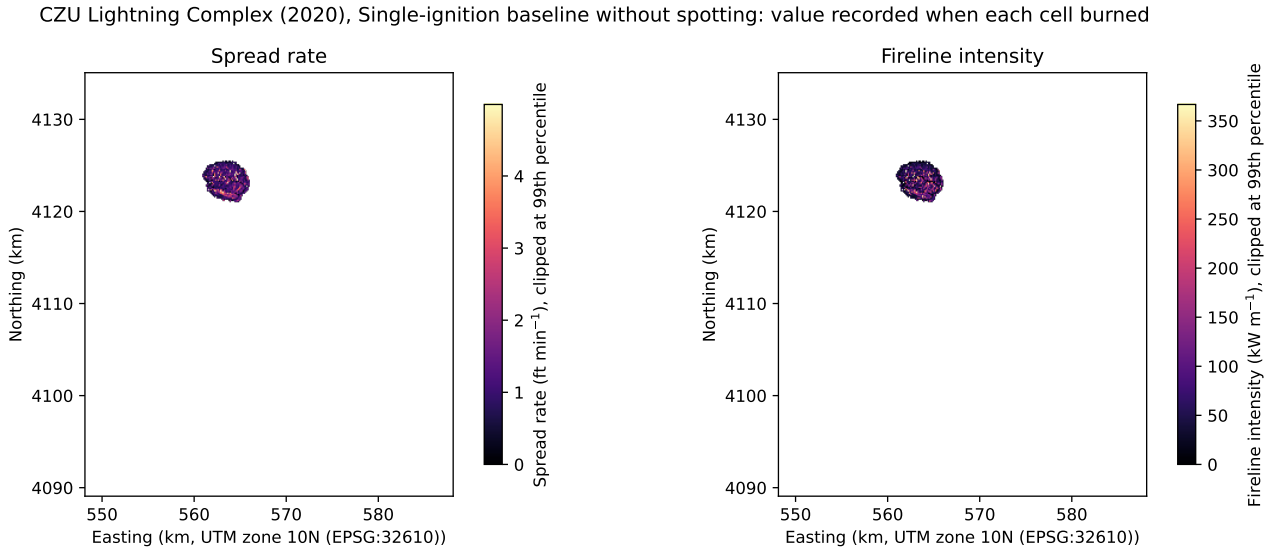



Figure 8: Single-ignition baseline without spotting: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

## Reference-case spotting configuration

This result used every GOFER ignition point at its observed time, the ELMFIRE default crown-fire treatment, and firebrand spotting with the settings of the landscape-scale reconstructions of the Camp, Thomas and Tubbs fires: firebrands generated in proportion to fire power from both surface and crown fire, an empirical transport-distance model, Eulerian accumulation and a physical ignition model. Inputs, stop time and numerical controls match the current configuration. The configuration file used was retained with this result. This result was produced before offshore cells outside LANDFIRE coverage were filled with open-water values; those cells then held a fill value read as data. They lie offshore beyond a band of open water that the simulated fire did not reach, but the result is not from the corrected

inputs.

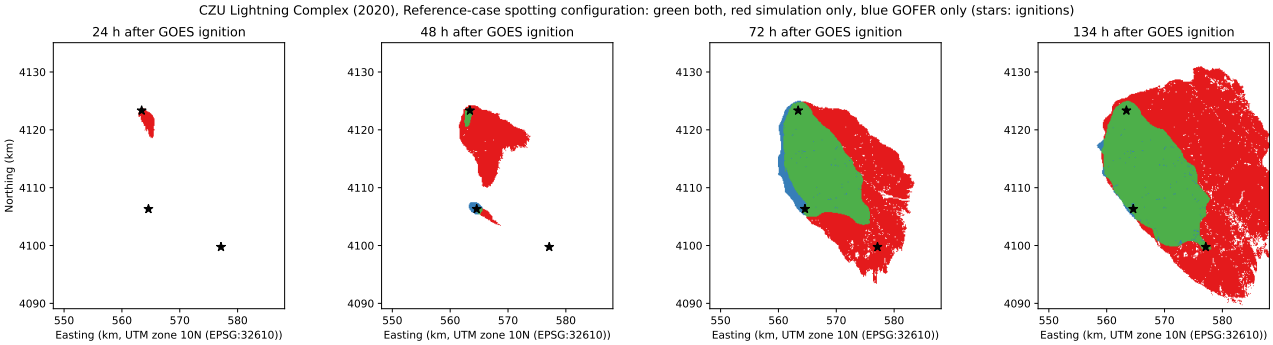



Figure 9: Reference-case spotting configuration: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

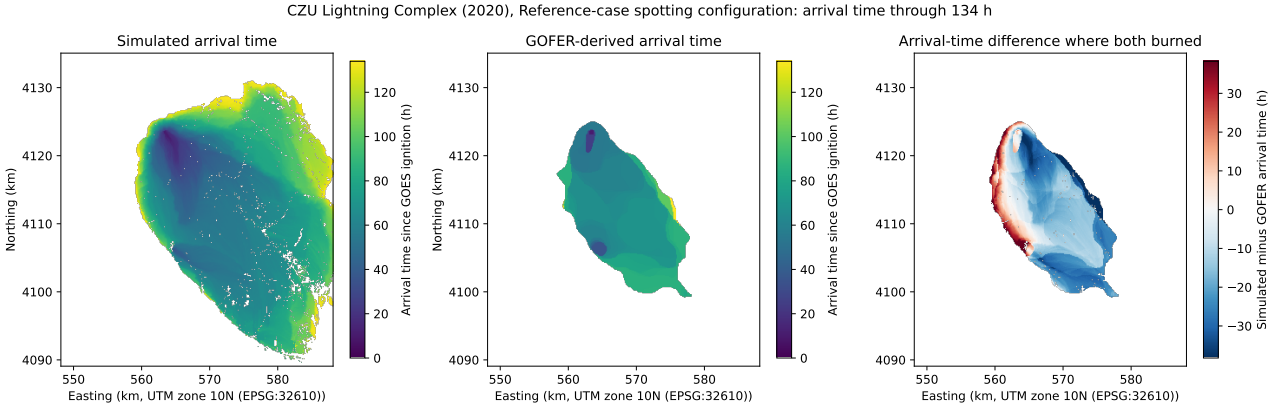



Figure 10: Reference-case spotting configuration: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

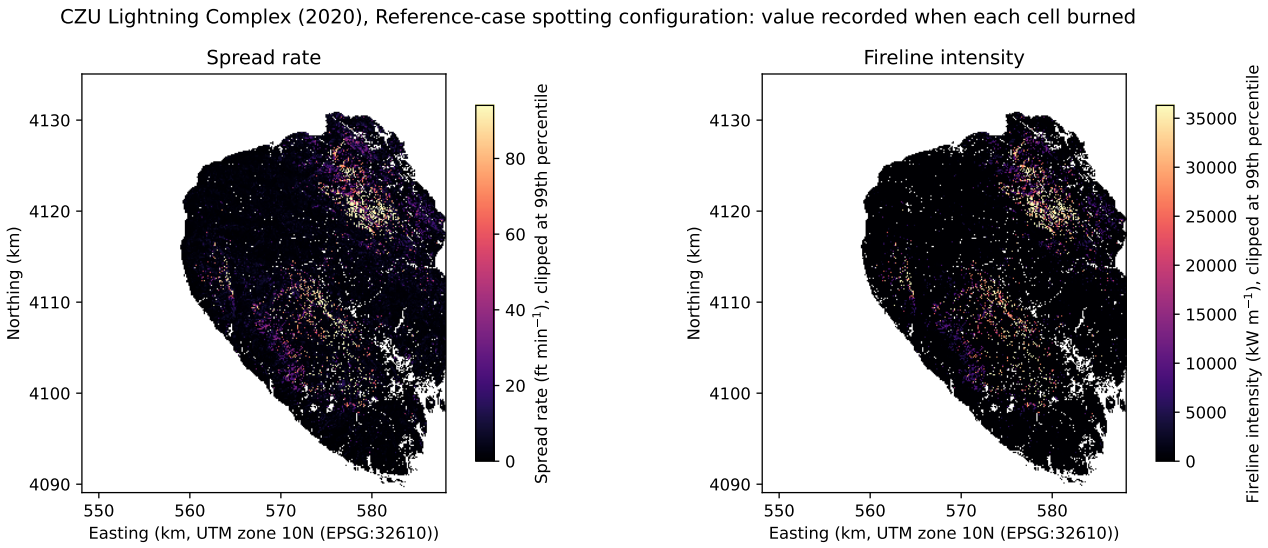



Figure 11: Reference-case spotting configuration: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

# 8 Integrated statistics and calculated metrics

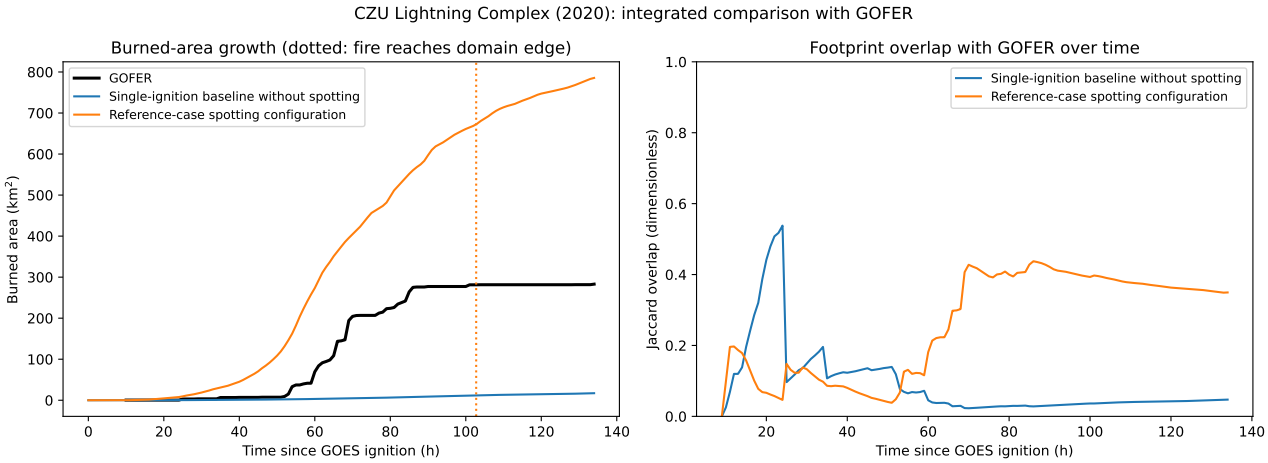



Figure 12: Observed (GOFER) and simulated burned-area growth, and the Jaccard overlap of each simulation with the GOFER footprint, evaluated hourly since the GOES ignition hour. Dotted lines mark the time each simulated fire first reaches the model-domain boundary.

## Single-ignition baseline without spotting

| Time (h) | GOFER (km$^2$) | Simulated (km$^2$) | Area ratio | Jaccard | Precision | Recall | $F_1$ |
|---|---|---|---|---|---|---|---|
| 24 | 0.391 | 0.266 | 0.68 | 0.54 | 0.86 | 0.59 | 0.7 |
| 48 | 7.38 | 1.71 | 0.232 | 0.13 | 0.63 | 0.15 | 0.24 |
| 72 | 207 | 5.01 | 0.0242 | 0.024 | 1 | 0.024 | 0.047 |
| 134 | 283 | 17.2 | 0.0607 | 0.047 | 0.79 | 0.048 | 0.09 |

Table 3: Footprint metrics of the single-ignition baseline without spotting against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

| Quantity | Value |
|---|---|
| Cells burned in both by 134 h | 15,037 |
| Mean arrival-time difference, simulated minus GOFER (h) | 33.2 |
| Median arrival-time difference (h) | 30.9 |
| Mean absolute arrival-time difference (h) | 34.6 |
| Root-mean-square arrival-time difference (h) | 42 |
| Simulated end since GOES ignition (h) | 134 |
| Propagation stalled before the planned end | no |
| First contact with domain boundary (h) | none |
| Final simulated area (ac) | 4,240 |
| Final crown-fire area (ac) | 9.3 |

Table 4: Arrival-time and integrated statistics of the single-ignition baseline without spotting. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Precision exceeds recall at the final comparison time, so the simulated footprint lies largely inside the observation but misses much of it: the simulation undergrows or follows a displaced trajectory. The simulated burned area is 0.061 times the observed area at 134 h. Where both burned, the simulated fire arrives on average 33.2 h later than the observed fire (mean absolute difference 34.6 h).

## Reference-case spotting configuration

| Time (h) | GOFER (km$^2$) | Simulated (km$^2$) | Area ratio | Jaccard | Precision | Recall | $F_1$ |
|---|---|---|---|---|---|---|---|
| 24 | 0.391 | 7.81 | 20 | 0.046 | 0.047 | 0.93 | 0.089 |
| 48 | 7.38 | 91.6 | 12.4 | 0.047 | 0.048 | 0.6 | 0.089 |
| 72 | 207 | 423 | 2.05 | 0.42 | 0.44 | 0.9 | 0.59 |
| 134 | 283 | 786 | 2.78 | 0.35 | 0.35 | 0.98 | 0.52 |

Table 5: Footprint metrics of the reference-case spotting configuration against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

| Quantity | Value |
|---|---|
| Cells burned in both by 134 h | 307,431 |
| Mean arrival-time difference, simulated minus GOFER (h) | -14 |
| Median arrival-time difference (h) | -15.7 |
| Mean absolute arrival-time difference (h) | 19.6 |
| Root-mean-square arrival-time difference (h) | 22.4 |
| Simulated end since GOES ignition (h) | 134 |
| Propagation stalled before the planned end | no |
| First contact with domain boundary (h) | 103 |
| Final simulated area (ac) | 193,156 |
| Final crown-fire area (ac) | 14,873 |

Table 6: Arrival-time and integrated statistics of the reference-case spotting configuration. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Recall exceeds precision at the final comparison time, so most of the observed footprint is reached but a larger share of the simulated footprint lies outside it: the simulation overgrows the observation. The simulated burned area is 2.8 times the observed area at 134 h. Where both burned, the simulated fire arrives on average 14.0 h earlier than the observed fire (mean absolute difference 19.6 h). The simulated fire reaches the model-domain boundary 103 h after the GOES ignition hour; growth after that time is truncated by the domain and later overlap values are affected.

## 9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics. The fire burned near the coast, where the marine layer and fog influenced late growth; the 2.5-km weather analysis may not resolve these effects.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

## 10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the CZU Lightning Complex. Inputs and the observation are characterized in full; the current scientific status is **CHARACTERIZED** with 2 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

## References


[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

# 2020 Glass Fire Reconstruction against GOFER Hourly Perimeters

## 1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Glass Fire (2020) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 189 h, from the first GOES fire detection at 27 September 2020 at 10:00 UTC to the last GOFER perimeter at 5 October 2020 at 07:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. No model parameter was tuned against the GOFER observation of this fire; the spotting configuration was revised after comparisons for the Kincade and CZU Lightning Complex fires.

## 2 Reference event, experiment, and observations

GOFER records the Glass Fire from 27 September 2020 at 10:00 UTC to 5 October 2020 at 07:00 UTC with 189 hourly perimeters, a final perimeter area of 353 km$^2$ and an official size of 67,484 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of $0.75 \pm 0.21$ km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 189 hourly perimeters of this fire on a 50-m grid in UTM zone 10N (EPSG:32610). The raster covers 352 km$^2$, consistent with the final GOFER perimeter area of 353 km$^2$; arrival times range from 1 to 189 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

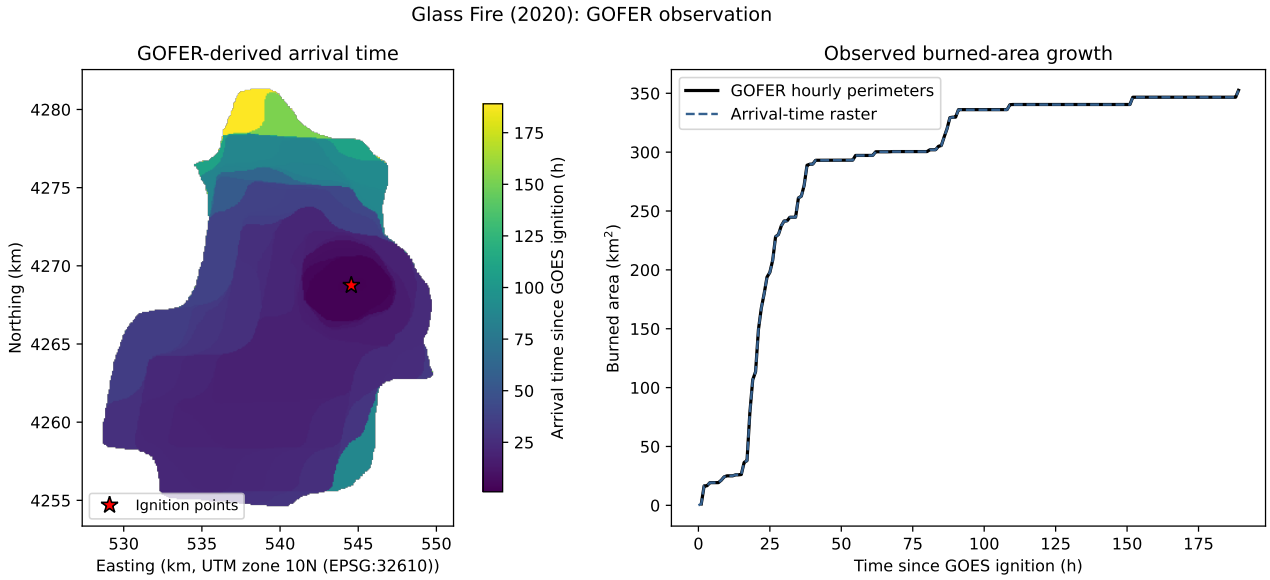



Figure 1: GOFER observation of the Glass Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

## 3 Input data provenance and characterization

| Input | Source and version | Resolution, units and processing |
|---|---|---|
| Terrain | LANDFIRE elevation, slope and aspect [3] | 30 m; m and degrees; UTM zone 10N (EPSG:32610) |
| Surface fuels | LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.1.0 | 30 m categorical |
| Canopy | LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.1.0 | 30 m; percent, m, m and kg m$^{-3}$ |
| Weather | Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6] | 600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees |
| Fuel moisture | Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather | 600 m hourly; percent; derivation method not documented in the available case materials |
| Ignition | GOFER ignition points [2] | Points with hourly time stamps |
| Adjustment and initial state | Spread-rate adjustment factor and initial level set | Uniform value of one (no adjustment, nothing burned at the start) |

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 1,400 by 1,600 cells of 30 m (2,016 km$^2$) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 14 to 1,318 m (median 275 m) and slope has a median of 12° and a 95th percentile of 29°; the circular-mean

aspect is 214°. By area the domain is 14.1% grass, 11.4% grass-shrub, 17.4% shrub, 16.9% timber-understory, 22% timber litter, 0.0231% slash-blowdown and 18.2% non-burnable fuel models. Canopy cover has a median of 0% and canopy base height a median of 0 m.

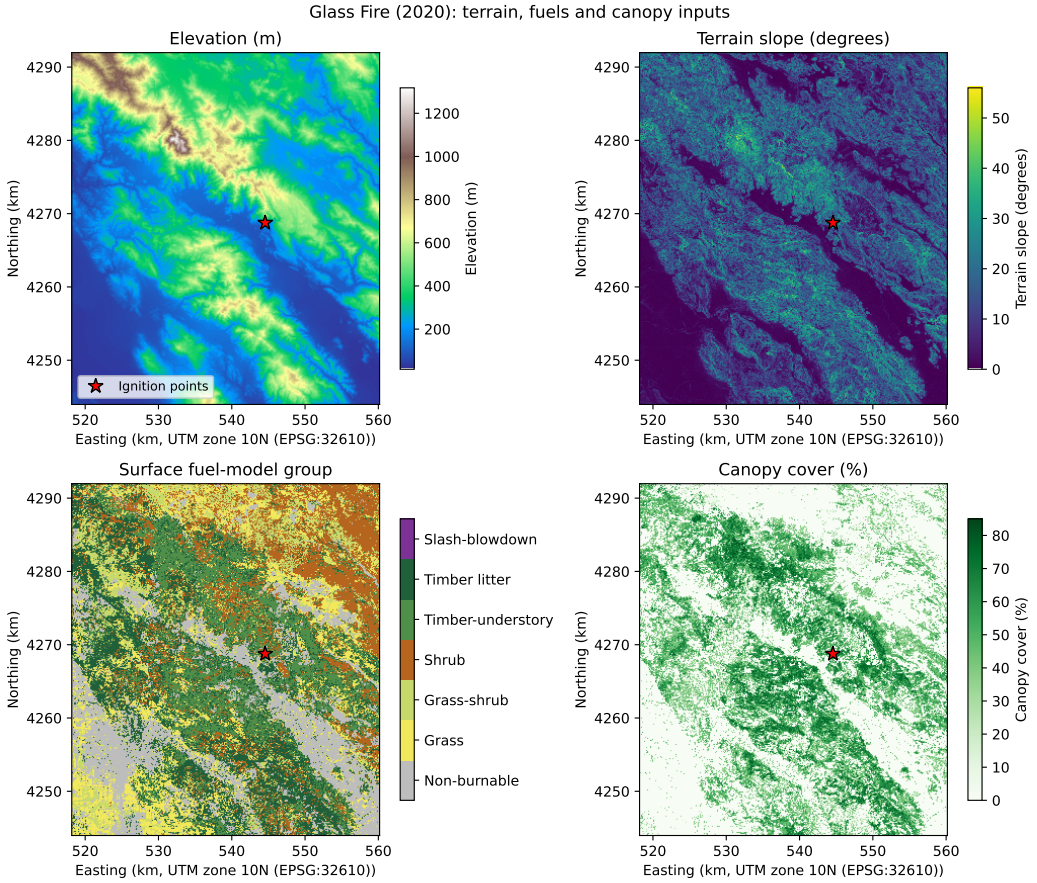



Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

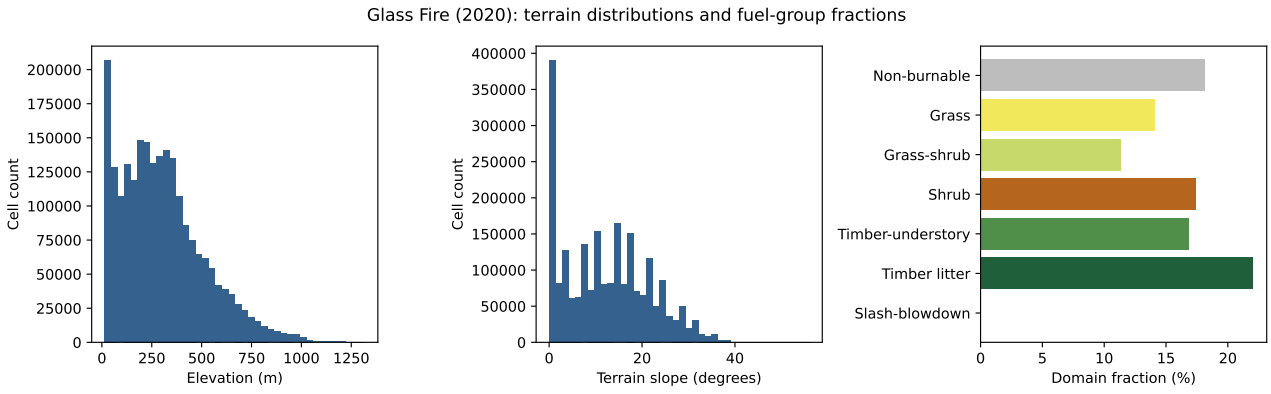



Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 190 hourly fields. Over all hours and cells the 20-ft wind speed has a median of 2.96 mph and a 95th percentile of 10.6 mph; the circular-mean direction the wind blows from is 322° with a mean resultant length of 0.194 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 5.61% (5th percentile 2.87%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

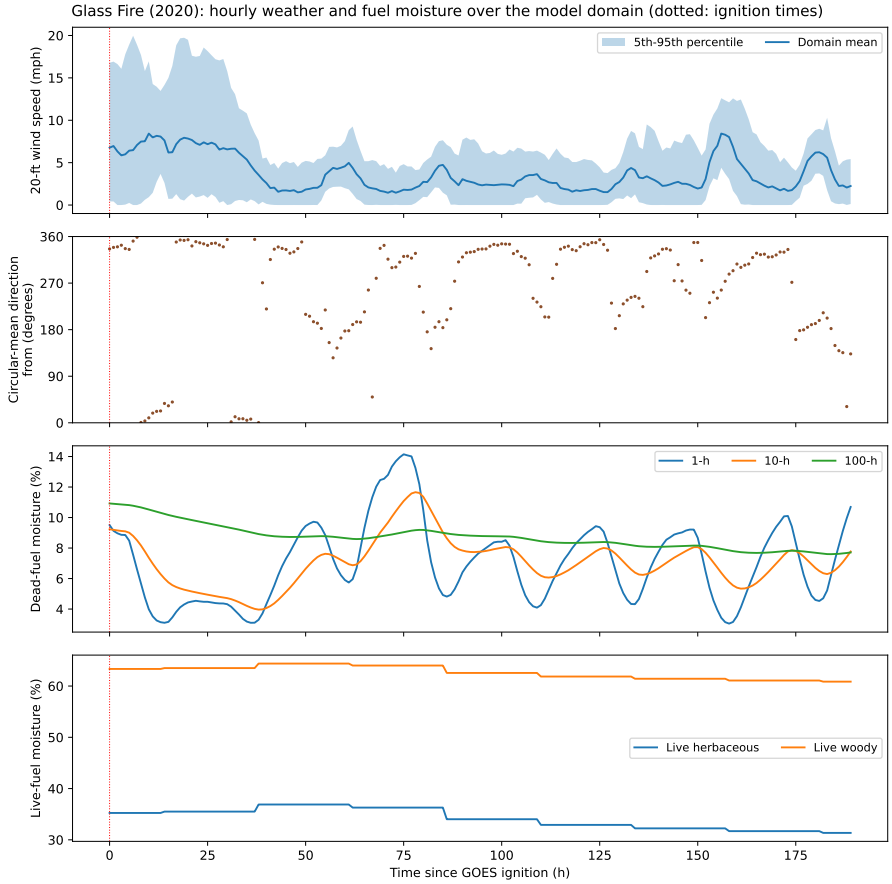



Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

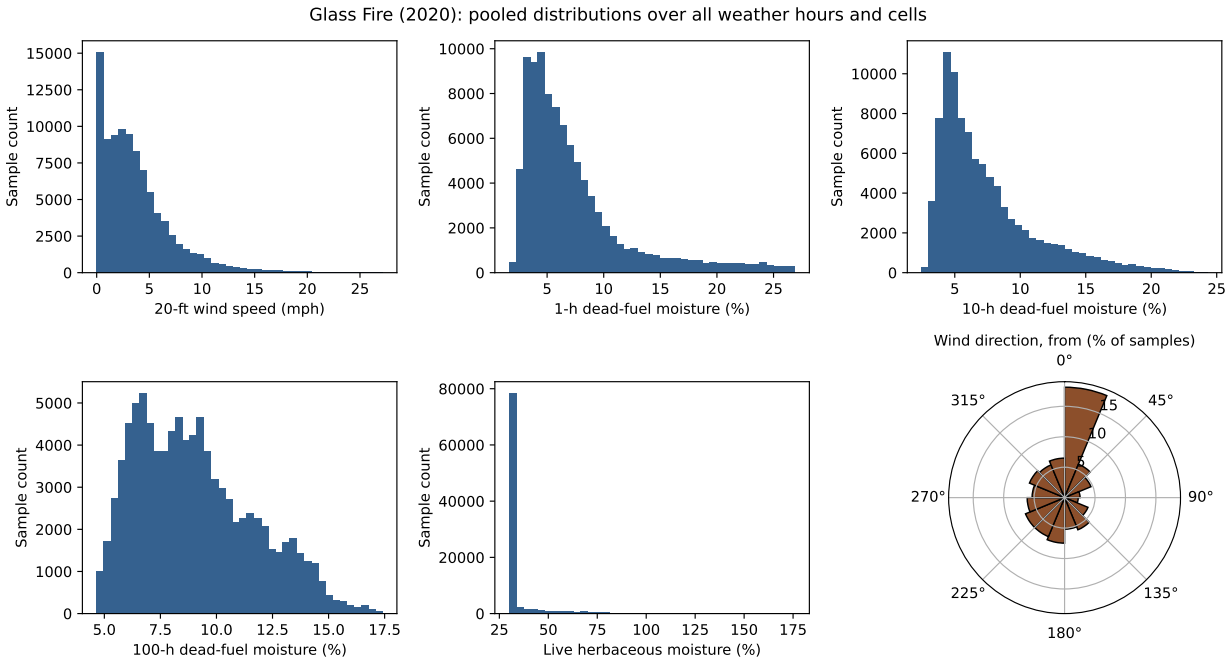



Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

## 4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 189 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft $\text{min}^{-1}$, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

| Choice | Basis | Evidence and limitation |
|---|---|---|
| Ignition points and times | documented | GOFER ignition record; no point required moving |
| Simulation start and stop | documented | First GOFER ignition point and last GOFER perimeter |
| Fuel vintage | derived | Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire |
| Weather and moisture | user-specified | RTMA analysis through CloudFire, as stated by the case owner |
| Crown fire | documented | ELMFIRE default crown-fire treatment, set explicitly |
| Firebrand spotting | user-specified | Enabled with ELMFIRE default spotting settings until calibration |
| Domain margin | user-specified | 10 km beyond the final GOFER perimeter |
| Single deterministic member | user-specified | No ensemble; stochastic spotting uses a fixed random seed |

Table 2: Basis of consequential configuration choices.

## 5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 0 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. The simulation results reported here were produced with ELMFIRE 1.1 at source revision `5ea7460d514b`, as recorded for the CZU Lightning Complex runs and assumed for the other results produced on the same system.

## 6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time $t$ the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap $J_t$ summarizes footprint agreement; precision $P_t$ diagnoses commission (simulated burning outside the observation), recall $R_t$ omission (observed burning the simulation misses), $F_t$ balances the two, and the area ratio $A_t$ exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

# 7 Simulation results and field-level comparison

The current status of this case is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Only results from earlier simulation configurations are available; the current case configuration has not been simulated.

## Single-ignition baseline without spotting

This result used a single point ignition at the first GOFER ignition point, firebrand spotting disabled and the ELMFIRE default crown-fire treatment, with the same inputs, stop time and numerical controls as the current configuration. The configuration file used was retained with this result.

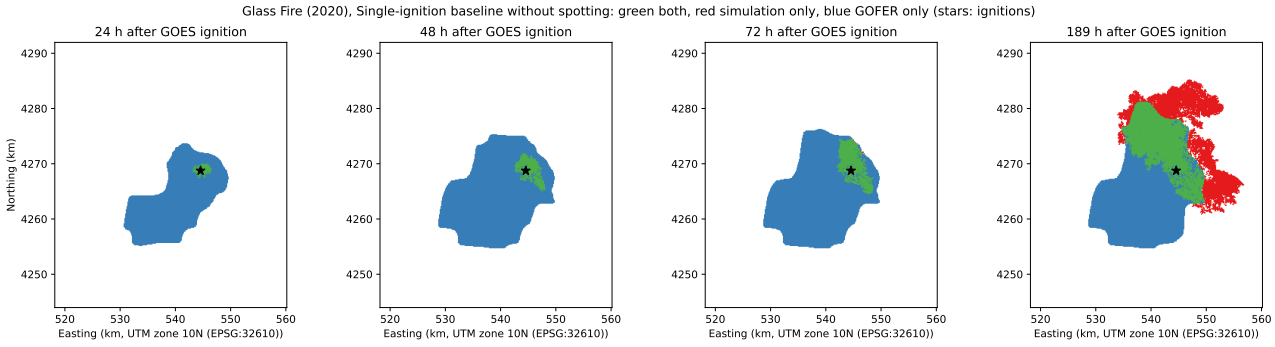



Figure 6: Single-ignition baseline without spotting: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

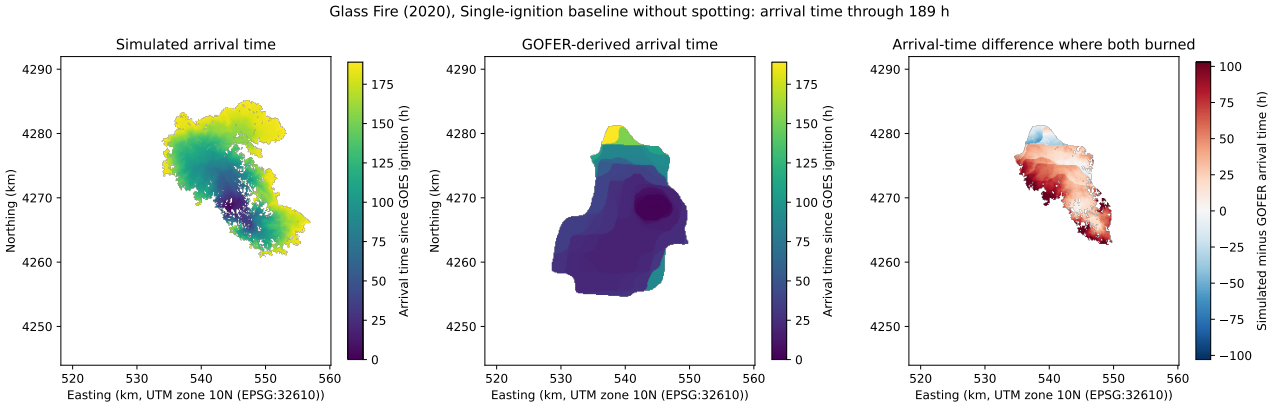



Figure 7: Single-ignition baseline without spotting: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

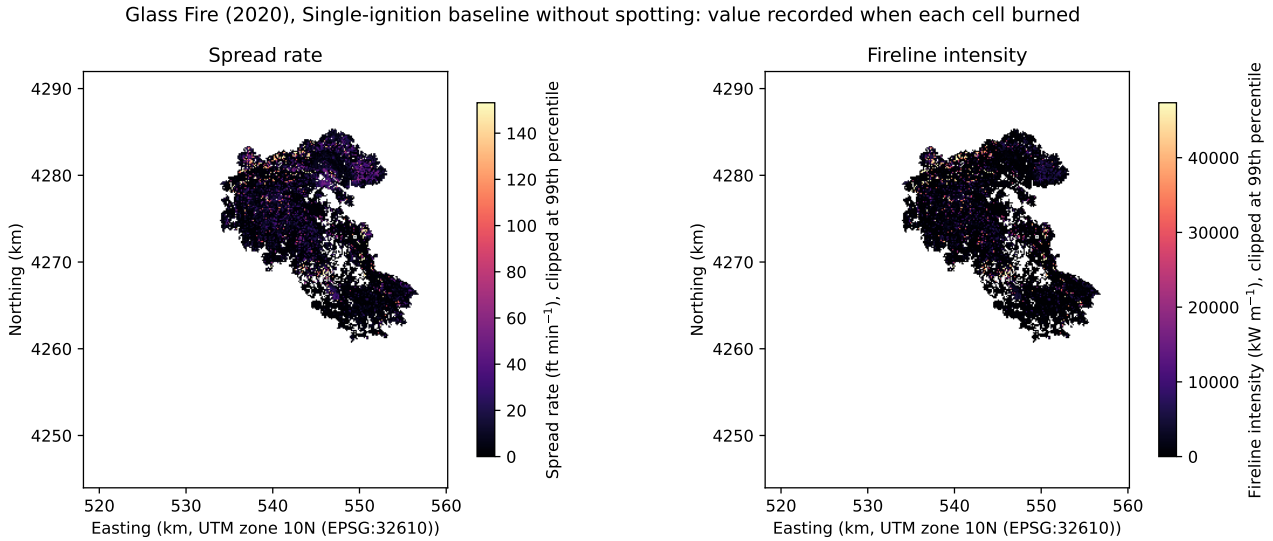



Figure 8: Single-ignition baseline without spotting: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

# 8 Integrated statistics and calculated metrics

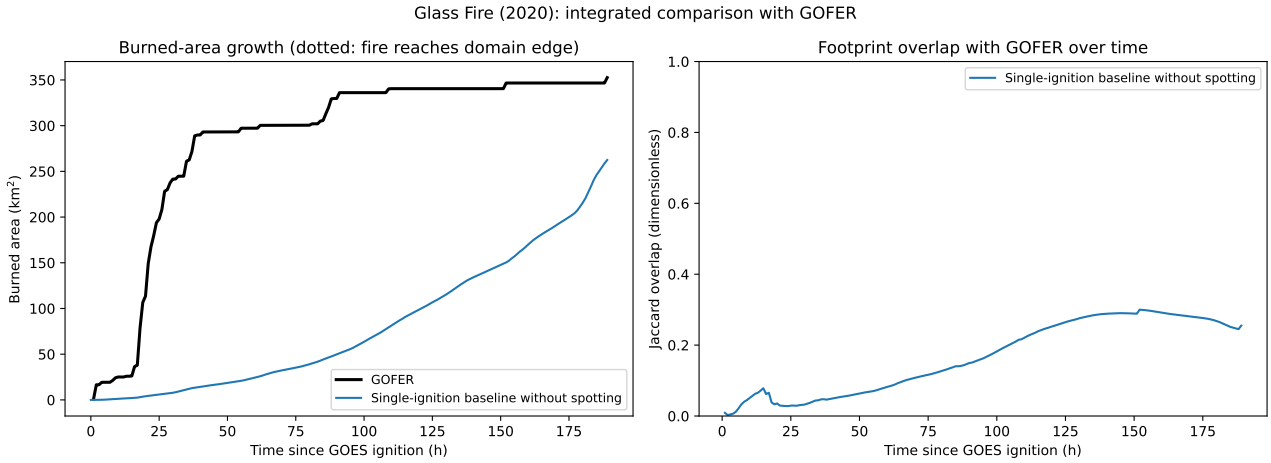



Figure 9: Observed (GOFER) and simulated burned-area growth, and the Jaccard overlap of each simulation with the GOFER footprint, evaluated hourly since the GOES ignition hour. Dotted lines mark the time each simulated fire first reaches the model-domain boundary.

## Single-ignition baseline without spotting

| Time (h) | GOFER (km$^2$) | Simulated (km$^2$) | Area ratio | Jaccard | Precision | Recall | $F_1$ |
|---|---|---|---|---|---|---|---|
| 24 | 194 | 5.4 | 0.0279 | 0.028 | 1 | 0.028 | 0.054 |
| 48 | 293 | 17.6 | 0.0602 | 0.06 | 1 | 0.06 | 0.11 |
| 72 | 300 | 33.6 | 0.112 | 0.11 | 1 | 0.11 | 0.2 |
| 189 | 352 | 262 | 0.745 | 0.25 | 0.48 | 0.35 | 0.41 |

Table 3: Footprint metrics of the single-ignition baseline without spotting against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

| Quantity | Value |
|---|---|
| Cells burned in both by 189 h | 138,800 |
| Mean arrival-time difference, simulated minus GOFER (h) | 41.7 |
| Median arrival-time difference (h) | 37.1 |
| Mean absolute arrival-time difference (h) | 46.5 |
| Root-mean-square arrival-time difference (h) | 55.4 |
| Simulated end since GOES ignition (h) | 189 |
| Propagation stalled before the planned end | no |
| First contact with domain boundary (h) | none |
| Final simulated area (ac) | 64,855 |
| Final crown-fire area (ac) | 13,320 |

Table 4: Arrival-time and integrated statistics of the single-ignition baseline without spotting. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Precision exceeds recall at the final comparison time, so the simulated footprint lies largely inside the observation but misses much of it: the simulation undergrows or follows a displaced trajectory. The simulated burned area is 0.74 times the observed area at 189 h. Where both burned, the simulated fire arrives on average 41.7 h later than the observed fire (mean absolute difference 46.5 h).

## 9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

## 10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Glass Fire. Inputs and the observation are characterized in full; the current scientific status is **CHARACTERIZED** with 1 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

## References


[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

[2] T. Liu et al., GOFER product and algorithm for 28 California fires, 2019–2021, Zenodo, 2023, doi:10.5281/zenodo.8327264.

[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

# 2019 Kincade Fire Reconstruction against GOFER Hourly Perimeters

## 1 Validation objective and scope

This case asks whether ELMFIRE, driven by LANDFIRE fuels and terrain, hourly analyzed weather and the observed ignition points, reproduces the hourly growth of the Kincade Fire (2019) recorded by the GOES-Observed Fire Event Representation (GOFER) product. The comparison spans 134 h, from the first GOES fire detection at 24 October 2019 at 04:00 UTC to the last GOFER perimeter at 29 October 2019 at 18:00 UTC. The prediction quantity is the ELMFIRE fire arrival time on the model grid; the reference quantity is the arrival time implied by the GOFER hourly perimeters. The case exercises the coupled surface-fire, crown-fire, firebrand-spotting and level-set spread processes of ELMFIRE. It does not isolate any single algorithm, and it does not represent fire suppression, which strongly affected the late growth of most of these fires.

The question is two-sided: does the simulated footprint reach the observed burned area, and does it do so without excessive growth outside it? Overlap, omission, commission, area and arrival-time error are therefore reported together, at fixed times after ignition. No independent, predeclared acceptance thresholds exist for this fire, so results are *characterized*; they are not labeled PASS or FAIL. The firebrand-spotting configuration was changed after the case owner inspected comparisons of this fire with GOFER, so results for this fire are not independent of that choice.

## 2 Reference event, experiment, and observations

GOFER records the Kincade Fire from 24 October 2019 at 04:00 UTC to 29 October 2019 at 18:00 UTC with 134 hourly perimeters, a final perimeter area of 347 km$^2$ and an official size of 77,758 ac, and 1 ignition point(s). GOFER derives hourly fire perimeters from the active-fire detections of the GOES-East and GOES-West geostationary satellites [1, 2]. For each hour the maximum detection confidence of each satellite over the roughly 10-minute full-disk images is combined, corrected for terrain parallax, smoothed, thresholded and vectorized into a cumulative perimeter. The combined product used here has an area-weighted grid resolution of about 1.7 km. Against California Fire and Resource Assessment Program final perimeters for the 28 fires of the dataset, the combined product reports a mean intersection over union of 0.77 and a mean perimeter-position error of $0.75 \pm 0.21$ km [1].

The observation used for comparison is an arrival-time raster in which each cell holds the first hour, measured from the GOES ignition hour, at which it lies inside a cumulative GOFER perimeter. It was produced by rasterizing the 134 hourly perimeters of this fire on a 50-m grid in UTM zone 10N (EPSG:32610). The raster covers 347 km$^2$, consistent with the final GOFER perimeter area of 347 km$^2$; arrival times range from 1 to 134 h. Because the satellite pixels are about 2 km across, the observed arrival time is uncertain by roughly one hour and the footprint edge by several hundred meters, and growth below the detection limit of GOES is not recorded. The early perimeters of GOFER are scaled to account for small fires that are not resolved [1]. These limitations apply equally to every simulation compared here.

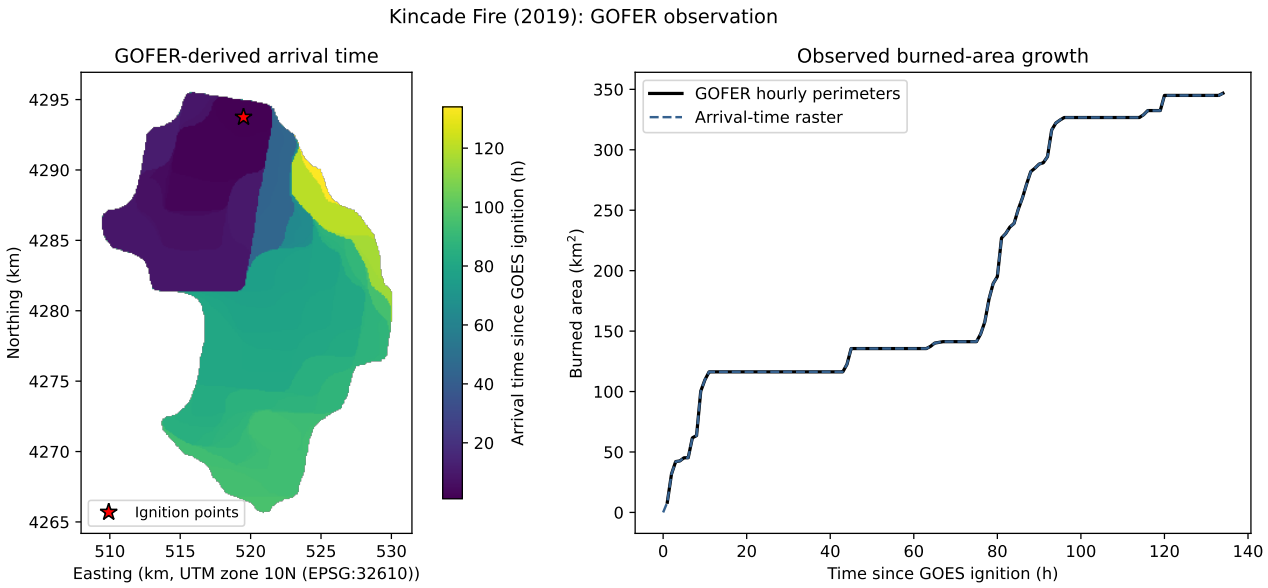



Figure 1: GOFER observation of the Kincade Fire: arrival time derived from the hourly perimeters (left; hours since the GOES ignition hour, UTM zone 10N (EPSG:32610)) and observed burned-area growth from the hourly perimeters and from the arrival-time raster (right). Stars mark the GOFER ignition points.

## 3 Input data provenance and characterization

| Input | Source and version | Resolution, units and processing |
|---|---|---|
| Terrain | LANDFIRE elevation, slope and aspect [3] | 30 m; m and degrees; UTM zone 10N (EPSG:32610) |
| Surface fuels | LANDFIRE 40 Scott and Burgan fuel models [4, 3], LANDFIRE release 2.0.0_2019 | 30 m categorical |
| Canopy | LANDFIRE canopy cover, height, base height and bulk density, LANDFIRE release 2.0.0_2019 | 30 m; percent, m, m and kg $m^{-3}$ |
| Weather | Real-Time Mesoscale Analysis (RTMA) [5], retrieved as hourly fields through the CloudFire service [6] | 600 m hourly; 20-ft wind speed in mph, direction the wind blows from in degrees |
| Fuel moisture | Dead (1-, 10- and 100-h) and live (herbaceous and woody) moisture derived by CloudFire from the analyzed weather | 600 m hourly; percent; derivation method not documented in the available case materials |
| Ignition | GOFER ignition points [2] | Points with hourly time stamps |
| Adjustment and initial state | Spread-rate adjustment factor and initial level set | Uniform value of one (no adjustment, nothing burned at the start) |

Table 1: Provenance of the model inputs. Fuels, terrain and weather were retrieved on 20 October 2026 for a domain centered on the GOFER perimeter.

The model domain is 1,400 by 1,667 cells of 30 m (2,100 $km^2$) in UTM zone 10N (EPSG:32610), extending 10 km beyond the bounding box of the final GOFER perimeter. Elevation ranges from 2 to 1,442 m (median 314 m) and slope has a median of 14° and a 95th percentile of 30°; the circular-mean

aspect is 201°. By area the domain is 16.3% grass, 7.73% grass-shrub, 12% shrub, 29.5% timber-understory, 18% timber litter, 0.137% slash-blowdown and 16.4% non-burnable fuel models. Canopy cover has a median of 35% and canopy base height a median of 0.3 m.

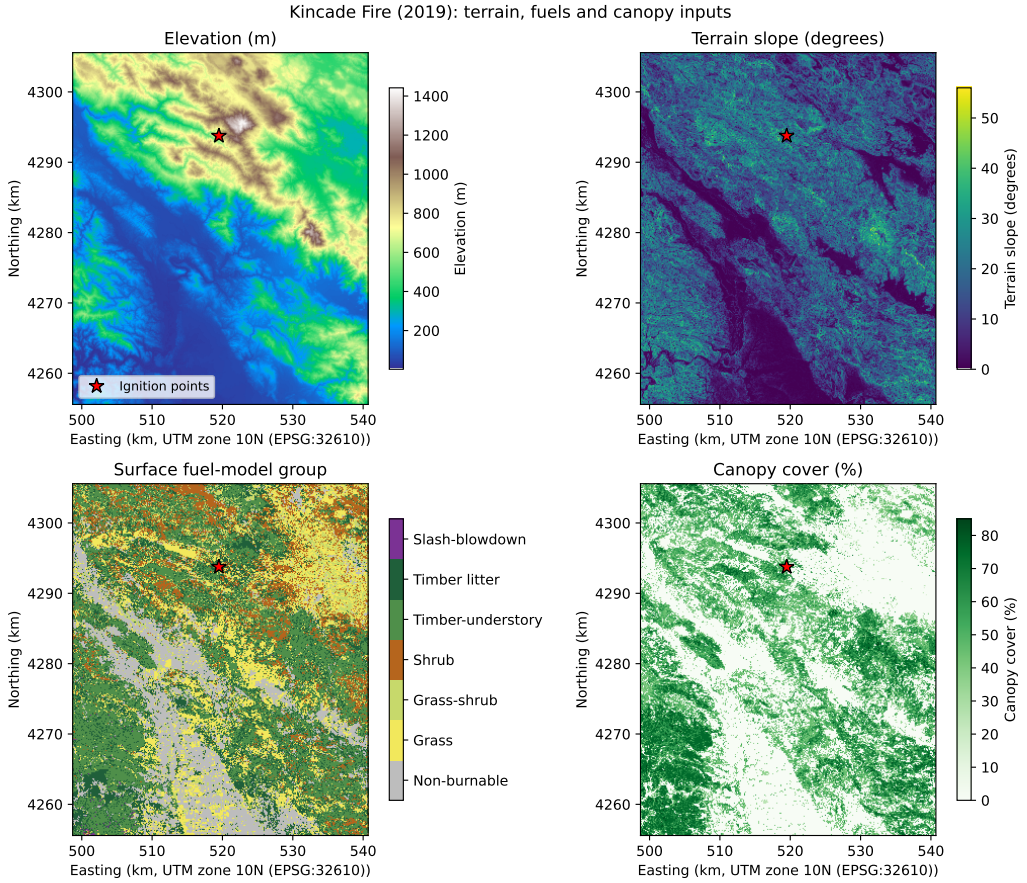



Figure 2: Model inputs over the whole domain: elevation, terrain slope, surface fuel-model group and canopy cover on the 30-m grid (UTM zone 10N (EPSG:32610)). Stars mark the ignition points. These fields characterize model forcing; they are not validation evidence.

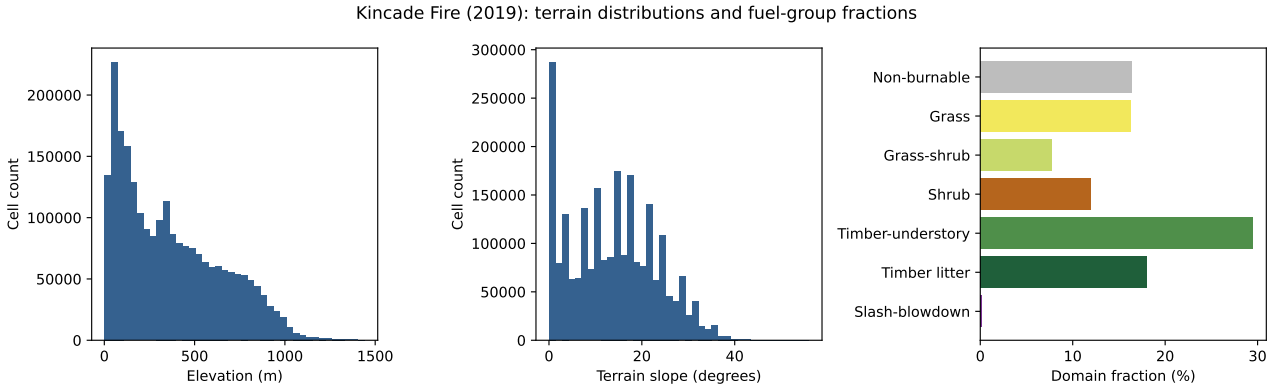



Figure 3: Distributions of elevation and terrain slope over valid cells, and fraction of the domain in each surface fuel-model group.

The weather comprises 135 hourly fields. Over all hours and cells the 20-ft wind speed has a median of 4.1 mph and a 95th percentile of 19.2 mph; the circular-mean direction the wind blows from is 1.42° with a mean resultant length of 0.309 (one for a constant direction, zero for uniformly spread directions). The 1-h dead-fuel moisture has a median of 4.57% (5th percentile 2.75%), and live herbaceous and woody moisture have medians of 30% and 60%. Pooled values hide the evolution of the event, so domain summaries are also shown by hour.

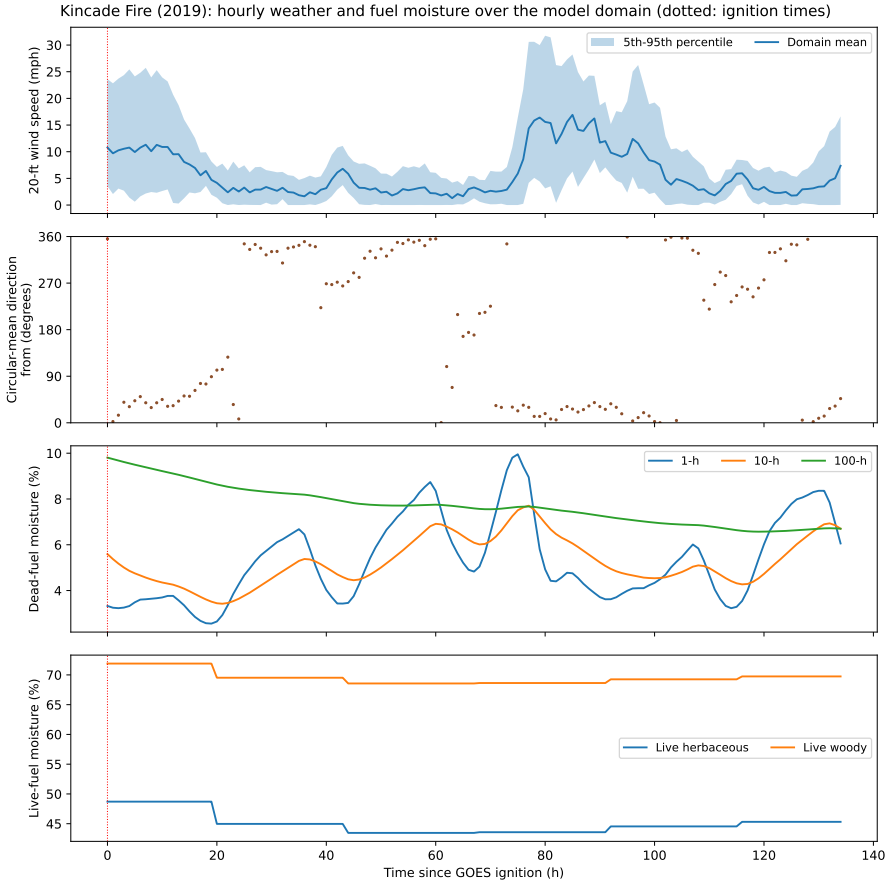



Figure 4: Hourly weather and fuel moisture over the model domain, in hours since the GOES ignition hour: domain-mean 20-ft wind speed with its 5th–95th percentile range, circular-mean wind direction (direction the wind blows from), and domain-mean dead- and live-fuel moisture. Dotted lines mark the ignition times.

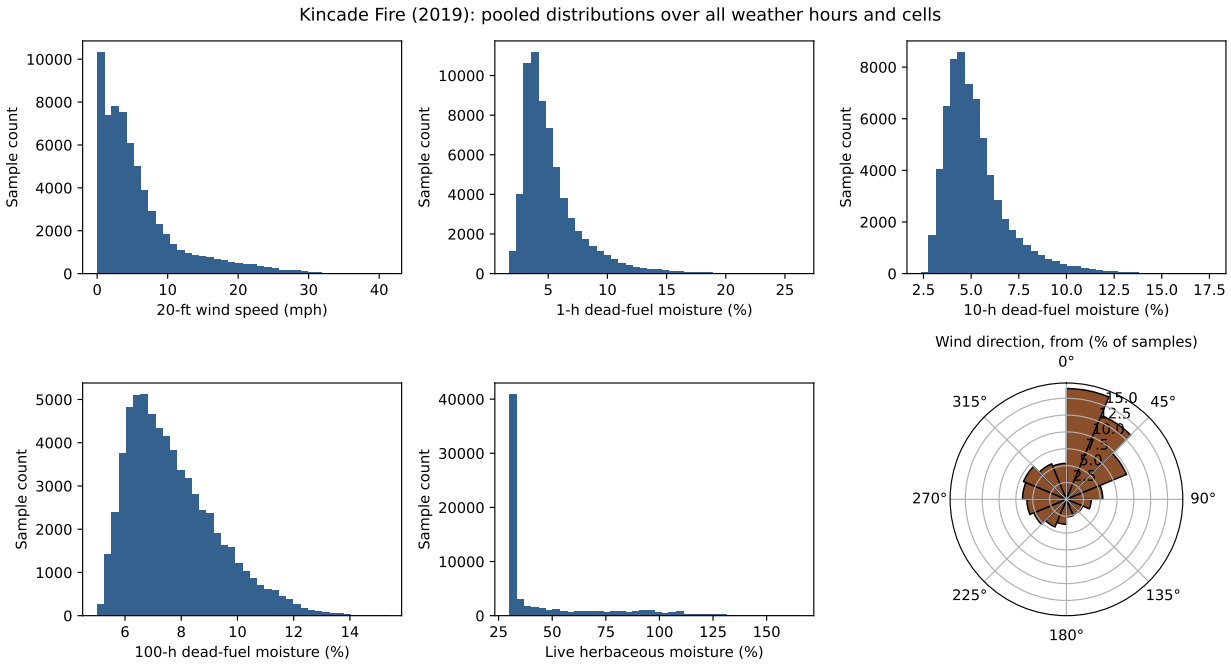



Figure 5: Pooled distributions of wind speed and fuel moisture over all hours and a deterministic regular sample of cells, and the wind-direction rose (fraction of samples by the direction the wind blows from).

## 4 Model configuration and rationale

The domain is the 30-m landscape grid in UTM zone 10N (EPSG:32610). The simulation starts at the first GOFER ignition point and runs 134 h, to the last GOFER perimeter, with a single point ignition. Weather and fuel moisture are read hourly and interpolated in time. Surface spread follows the Rothermel model with the Scott and Burgan fuel models. Crown fire uses the ELMFIRE default treatment: initiation when surface fireline intensity exceeds the critical intensity of Van Wagner [7], and active-crown spread rate from Cruz et al. [8], capped at 250 ft $\text{min}^{-1}$, with a constant foliar moisture of 90%. Firebrand spotting is enabled with every other spotting setting at its ELMFIRE default, so firebrands are generated only by crown-fire cells. The level-set equation is advanced with an adaptive timestep targeting a Courant–Friedrichs–Lewy (CFL) number of 0.2, starting at 1 s and limited to 600 s. The run is a single deterministic member with a fixed random seed; arrival time, spread rate, fireline intensity and fire-size statistics are recorded at the stop time.

| Choice | Basis | Evidence and limitation |
|---|---|---|
| Ignition points and times | documented | GOFER ignition record; no point required moving |
| Simulation start and stop | documented | First GOFER ignition point and last GOFER perimeter |
| Fuel vintage | derived | Latest LANDFIRE release whose mapped disturbances end before the fire year, so the fuels do not contain this fire |
| Weather and moisture | user-specified | RTMA analysis through CloudFire, as stated by the case owner |
| Crown fire | documented | ELMFIRE default crown-fire treatment, set explicitly |
| Firebrand spotting | user-specified | Enabled with ELMFIRE default spotting settings until calibration |
| Domain margin | user-specified | 10 km beyond the final GOFER perimeter |
| Single deterministic member | user-specified | No ensemble; stochastic spotting uses a fixed random seed |

Table 2: Basis of consequential configuration choices.

## 5 Preprocessing and reproducibility

Fuels, terrain and weather were retrieved through the CloudFire service for a rectangular domain centered on the bounding box of the final GOFER perimeter. The eight terrain, fuel and canopy layers were stacked into one landscape raster without resampling. Offshore cells outside LANDFIRE coverage, delivered with a fill value that is not the declared missing-data value, were assigned LANDFIRE's own open-water values (non-burnable fuel, zero elevation, slope and canopy). Each GOFER ignition point was projected to UTM zone 10N (EPSG:32610); a point on a non-burnable cell was moved to the nearest burnable cell within 300 m. Simulation time starts at the first ignition hour, 0 h after the GOES ignition hour; all results are reported in hours since the GOES ignition hour.

The provided inputs are checked before analysis: required files must be present with the recorded sizes, the landscape must have eight bands in the recorded coordinate reference system and no unfilled cells, and every weather field must cover the landscape with the recorded number of hourly bands. The input statistics and figures are then regenerated. Large inputs and simulation results are supplied to the case rather than stored in it; the source record identifies each file, its size and its role. The simulation results reported here were produced with ELMFIRE 1.1 at source revision `5ea7460d514b`, as recorded for the CZU Lightning Complex runs and assumed for the other results produced on the same system.

## 6 Validation metrics and comparison method

Let $\tau_s(\mathbf{x})$ be the simulated and $\tau_o(\mathbf{x})$ the GOFER arrival time of cell $\mathbf{x}$, both in hours since the GOES ignition hour. The simulated arrival time is the time-of-arrival output of the single simulation member at its final output time, shifted by the offset between the simulation start and the GOES ignition hour; unburned cells have no arrival time. The GOFER raster is transferred to the 30-m model grid by nearest-neighbor sampling, so observed times are never interpolated, and the comparison region is every valid landscape cell. At comparison time $t$ the footprints are $S_t = \{\mathbf{x} : \tau_s \le t\}$ and $O_t = \{\mathbf{x} : \tau_o \le t\}$, and with $I_t = |S_t \cap O_t|$,

$$J_t = \frac{I_t}{|S_t \cup O_t|}, \qquad P_t = \frac{I_t}{|S_t|}, \qquad R_t = \frac{I_t}{|O_t|}, \qquad F_t = \frac{2P_tR_t}{P_t + R_t}, \qquad A_t = \frac{|S_t|}{|O_t|}. \quad (1)$$

The Jaccard overlap $J_t$ summarizes footprint agreement; precision $P_t$ diagnoses commission (simulated burning outside the observation), recall $R_t$ omission (observed burning the simulation misses), $F_t$ balances the two, and the area ratio $A_t$ exposes size compensation that overlap alone can hide. The fixed comparison times are 24, 48 and 72 h and the last GOFER arrival time; fixed times avoid choosing favorable instants, and the early times limit the influence of suppression, which the model does not represent. For cells burned in both footprints by the final time, the arrival-time difference $\tau_s - \tau_o$ is summarized by its mean (bias), median, mean absolute value and root mean square. Burned-area growth and hourly Jaccard overlap give the time history. The time at which the simulated fire first reaches the domain boundary and whether propagation stalled before the planned end are reported because both truncate simulated growth.

Results from different simulation configurations are evaluated separately and never pooled. A configuration is characterized when its time-of-arrival result is available; a case with no simulation result is NOT EVALUABLE. Because no acceptance threshold has been declared independently of these results, no result is labeled PASS or FAIL.

## 7 Simulation results and field-level comparison

The current status of this case is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Only results from earlier simulation configurations are available; the current case configuration has not been simulated.

## Reference-case spotting configuration

This result used every GOFER ignition point at its observed time, the ELMFIRE default crown-fire treatment, and firebrand spotting with the settings of the landscape-scale reconstructions of the Camp, Thomas and Tubbs fires: firebrands generated in proportion to fire power from both surface and crown fire, an empirical transport-distance model, Eulerian accumulation and a physical ignition model. Inputs, stop time and numerical controls match the current configuration. The configuration file was not retained with this result; the configuration is documented from the case owner's preparation record.

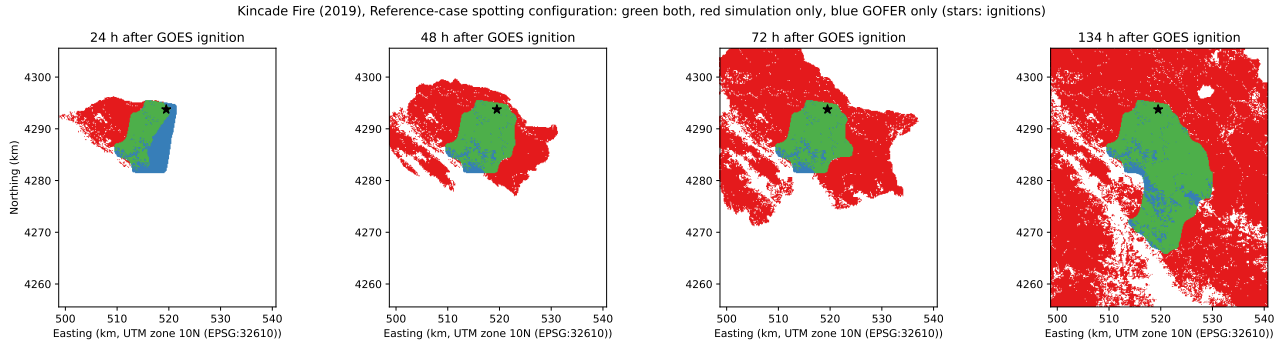



Figure 6: Reference-case spotting configuration: simulated and GOFER-derived footprints at the fixed comparison times (hours since the GOES ignition hour) on the 30-m model grid: cells burned in both (green), only in the simulation (red), and only in the GOFER observation (blue). Stars mark the configured ignition points.

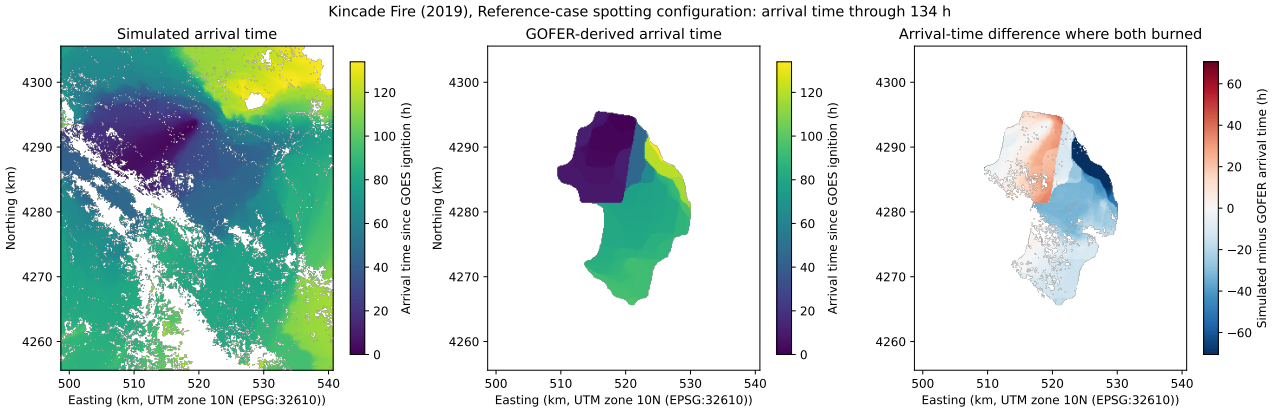



Figure 7: Reference-case spotting configuration: simulated arrival time, GOFER-derived arrival time and their difference where both burned through the final comparison time, all in hours since the GOES ignition hour. The difference color scale is symmetric and clipped at the 95th percentile of its magnitude.

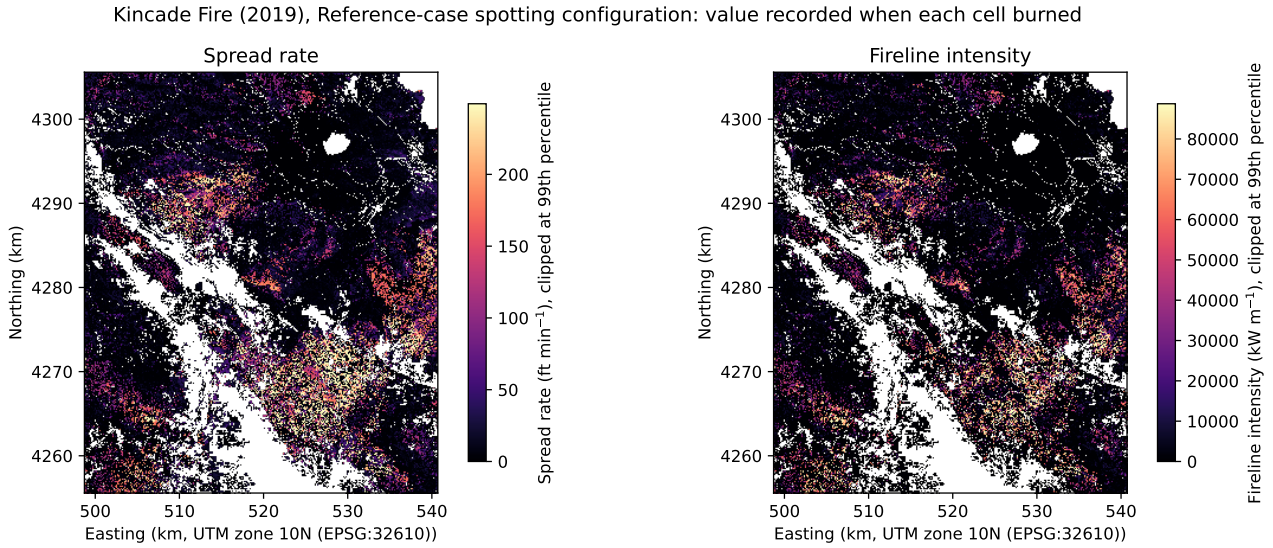



Figure 8: Reference-case spotting configuration: simulated spread rate and fireline intensity recorded when each cell burned, for the single simulation member at its final output time; color scales are clipped at the 99th percentile.

# 8 Integrated statistics and calculated metrics

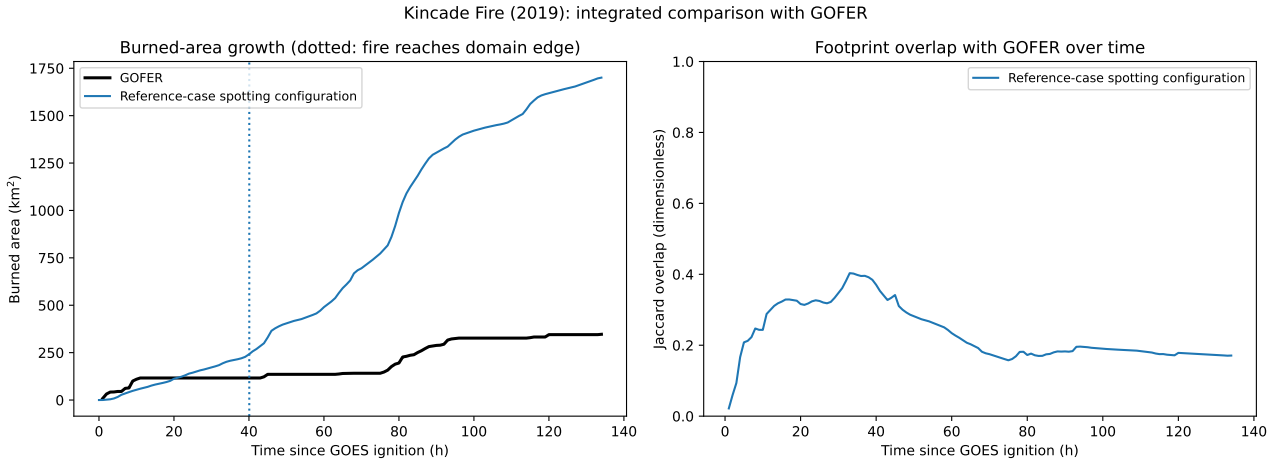



Figure 9: Observed (GOFER) and simulated burned-area growth, and the Jaccard overlap of each simulation with the GOFER footprint, evaluated hourly since the GOES ignition hour. Dotted lines mark the time each simulated fire first reaches the model-domain boundary.

## Reference-case spotting configuration

| Time (h) | GOFER (km$^2$) | Simulated (km$^2$) | Area ratio | Jaccard | Precision | Recall | $F_1$ |
|---|---|---|---|---|---|---|---|
| 24 | 116 | 139 | 1.2 | 0.33 | 0.45 | 0.54 | 0.49 |
| 48 | 136 | 389 | 2.87 | 0.29 | 0.31 | 0.88 | 0.45 |
| 72 | 141 | 724 | 5.13 | 0.17 | 0.17 | 0.88 | 0.29 |
| 134 | 347 | 1,701 | 4.9 | 0.17 | 0.18 | 0.86 | 0.29 |

Table 3: Footprint metrics of the reference-case spotting configuration against the GOFER-derived footprint at fixed times since the GOES ignition hour, evaluated on the 30-m model grid over valid landscape cells. Time is the comparison instant; areas are burned areas through that instant.

| Quantity | Value |
|---|---|
| Cells burned in both by 134 h | 332,015 |
| Mean arrival-time difference, simulated minus GOFER (h) | -12.1 |
| Median arrival-time difference (h) | -10.7 |
| Mean absolute arrival-time difference (h) | 20.9 |
| Root-mean-square arrival-time difference (h) | 27.9 |
| Simulated end since GOES ignition (h) | 134 |
| Propagation stalled before the planned end | no |
| First contact with domain boundary (h) | 40.1 |
| Final simulated area (ac) | 416,395 |
| Final crown-fire area (ac) | 104,650 |

Table 4: Arrival-time and integrated statistics of the reference-case spotting configuration. Arrival-time differences use cells burned in both the simulation and the GOFER observation by the final comparison time.

Recall exceeds precision at the final comparison time, so most of the observed footprint is reached but a larger share of the simulated footprint lies outside it: the simulation overgrows the observation. The simulated burned area is 4.9 times the observed area at 134 h. Where both burned, the simulated fire arrives on average 12.1 h earlier than the observed fire (mean absolute difference 20.9 h). The simulated fire reaches the model-domain boundary 40 h after the GOES ignition hour; growth after that time is truncated by the domain and later overlap values are affected.

## 9 Interpretation and uncertainty

Low overlap with high recall and low precision indicates overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory, and an area ratio near one cannot by itself establish spatial agreement. Late comparison times are dominated by processes outside the model: suppression, which stopped growth of the observed fire, and the finite model domain, which stops simulated growth at its boundary. The early comparison times are therefore the most informative about spread physics.

The dominant uncertainties are the coarse spatial resolution of the satellite observation and its hourly sampling, the 2.5-km resolution of the weather analysis, which smooths local terrain-driven winds, the LANDFIRE fuel classification, the undocumented derivation of fuel moisture, the ignition location uncertainty of about one GOES pixel, and the default firebrand-spotting parameters. A single deterministic member does not quantify uncertainty; model-form, parameter and input errors are not separable with these metrics alone.

## 10 Conclusions and limitations

The case characterizes the agreement between ELMFIRE and the GOFER hourly progression of the Kincade Fire. Inputs and the observation are characterized in full; the current scientific status is **CHARACTERIZED** with 1 simulation result(s) evaluated. The results describe the configurations listed and do not transfer to other configurations, fires or weather products. A defensible validation decision requires acceptance thresholds fixed before the evaluated run, an ensemble that represents weather, fuel and ignition uncertainty, documentation of the fuel-moisture derivation, and evaluation of any calibrated parameters on fires not used to choose them.

## References


[1] T. Liu, J. T. Randerson, Y. Chen, D. C. Morton, E. B. Wiggins, P. Smyth, E. Foufoula-Georgiou, R. Nadler and O. Nevo, "Systematically tracking the hourly progression of large wildfires using GOES satellite observations," *Earth System Science Data*, 16, 1395–1424, 2024, doi:10.5194/essd-16-1395-2024.

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[3] LANDFIRE, U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, landfire.gov.

[4] J. H. Scott and R. E. Burgan, "Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model," USDA Forest Service General Technical Report RMRS-GTR-153, 2005.

[5] M. S. F. V. De Pondeca et al., "The Real-Time Mesoscale Analysis at NOAA's National Centers for Environmental Prediction: current status and development," *Weather and Forecasting*, 26, 593–612, 2011.

[6] C. Lautenberger, ELMFIRE and the CloudFire fuel, weather and ignition data service, elmfire.io.

[7] C. E. Van Wagner, "Conditions for the start and spread of crown fire," *Canadian Journal of Forest Research*, 7, 23–34, 1977.

[8] M. G. Cruz, M. E. Alexander and R. H. Wakimoto, "Development and testing of models for predicting crown fire rate of spread in conifer forest stands," *Canadian Journal of Forest Research*, 35, 1626–1639, 2005.

# 2017 Thomas Fire Landscape Reconstruction

## 1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2017 Thomas Fire Landscape Reconstruction through 17 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

## 2 Reference event, experiment, and observations

The configured event start, expressed in Pacific standard time (PST), is 4 December 2017 at 18:30 PST. The comparison instant is 5 December 2017 at 11:30 PST (19:30 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 6804 point records in the geographic coordinate system defined by EPSG:4326; 1446 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

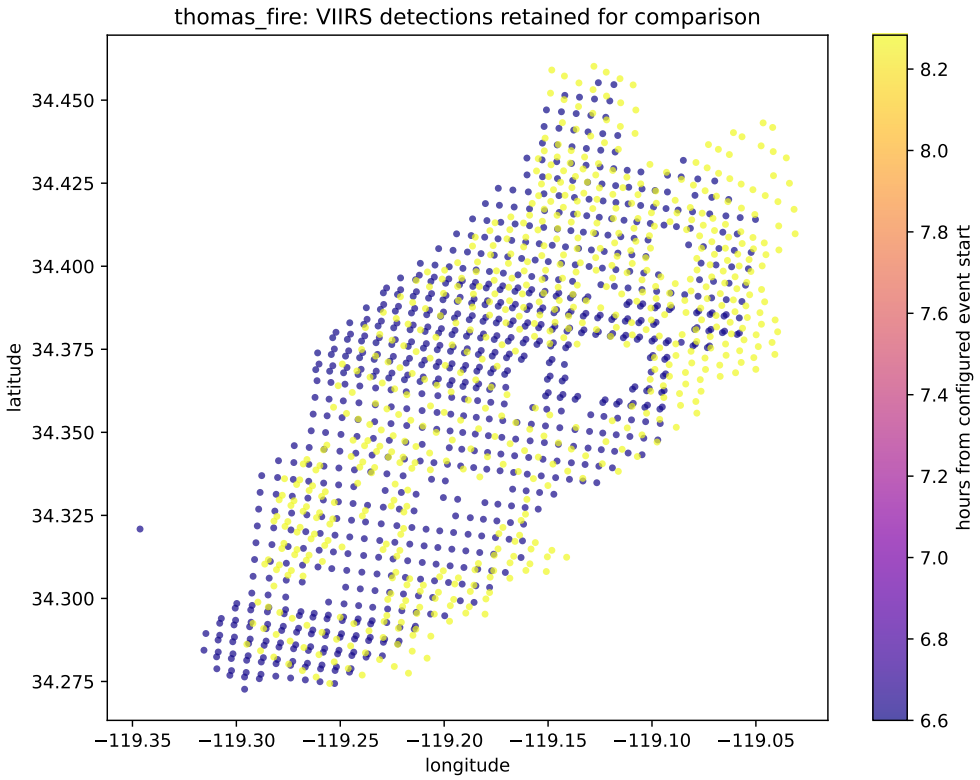



Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

## 3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `642689c45406337886a8dea52e08ff413fd3e7c2a933dcc6e0f7f07954a3a044`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

| Field | Minimum | Mean | Standard deviation | Maximum |
|---|---|---|---|---|
| Elevation | -1e+04 | -1007 | 3832 | 2286 |
| Slope | -1e+04 | -1502 | 3591 | 68 |
| Wind speed | 0 | 26.46 | 12.74 | 62.97 |
| 1-h moisture | 1.301 | 5.103 | 2.91 | 38.25 |
| 10-h moisture | 2.696 | 6.503 | 2.641 | 15.76 |
| 100-h moisture | 5.704 | 10.52 | 4.272 | 20.31 |

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

The active elevation raster contains values of -10000 even though its declared nodata value is -9999. The statistics retain those cells exactly as ELMFIRE sees them; the source owner must determine whether -10000 is a valid value or an undocumented sentinel before scientific interpretation.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 44.71° and 48.16°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most $512 \times 512$ cells per band, and are labeled accordingly.

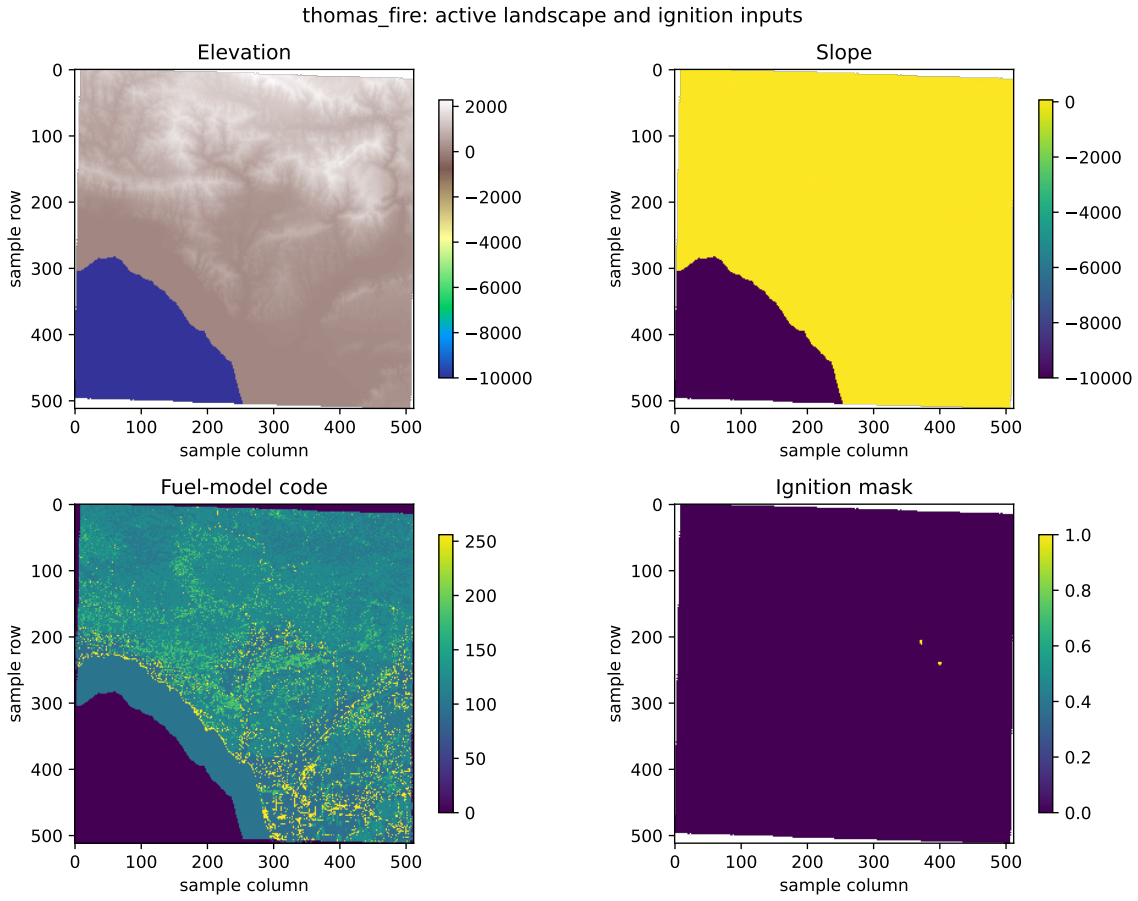



Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

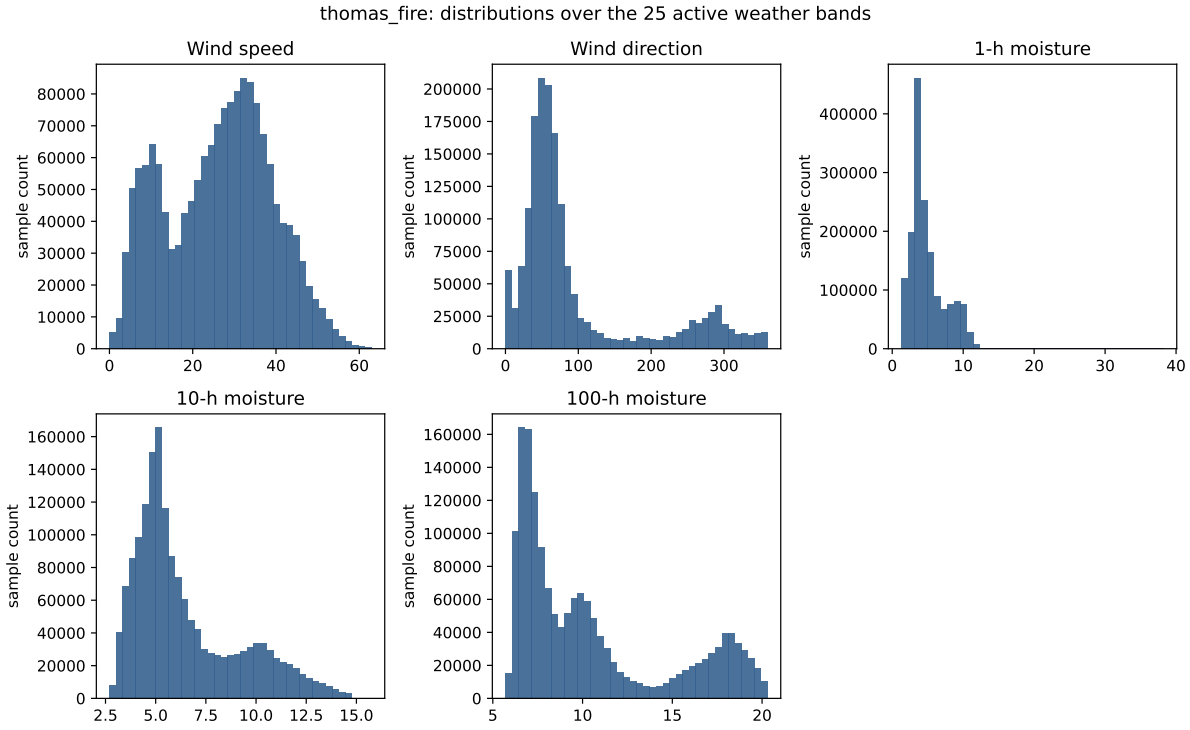



Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

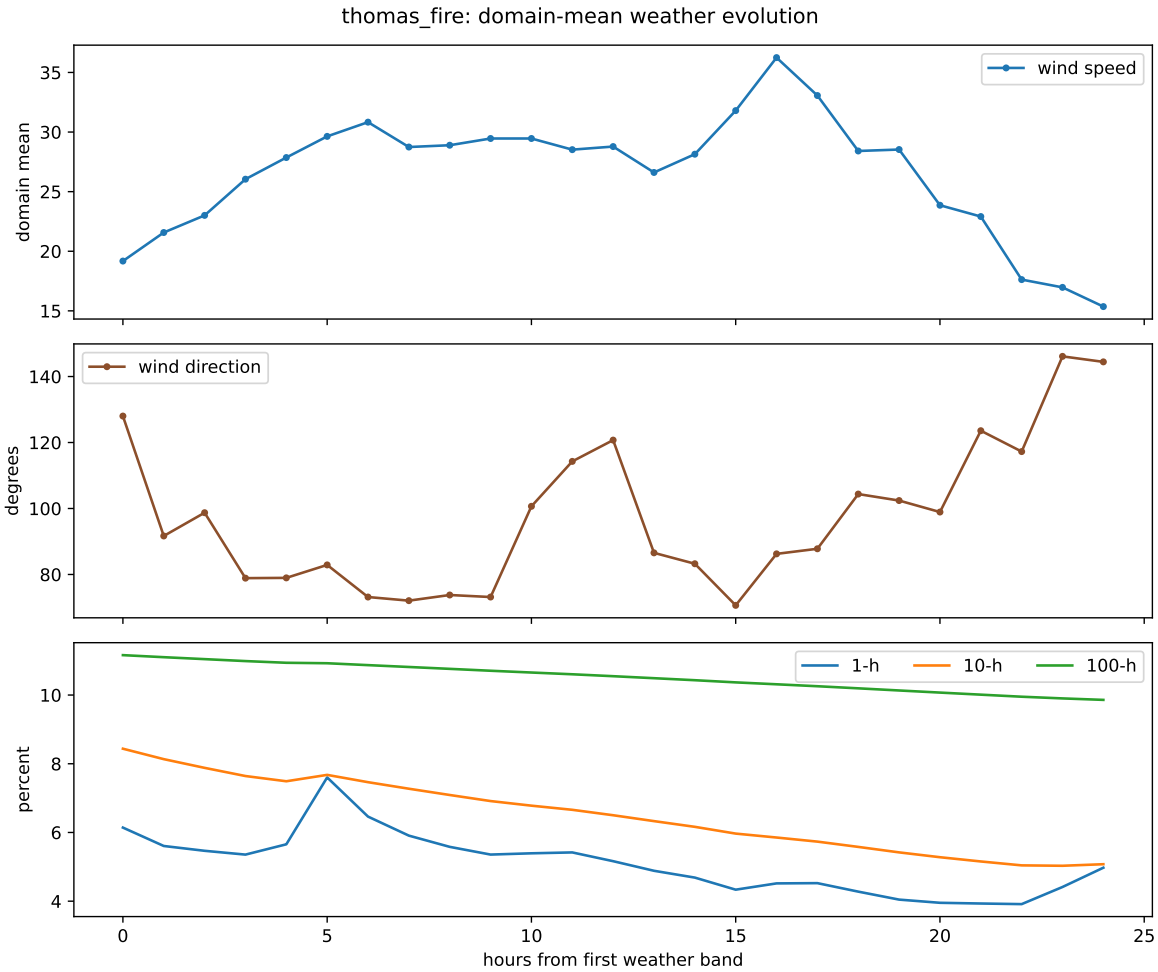



Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

## 4 Model configuration and rationale

The active domain is 2491 by 1779 cells at 30 m in EPSG:26911. Weather is 29-band weather rasters; bands 1–25 are selected at one-hour intervals, and the ignition treatment is two prescribed point ignitions at archived coordinates and times. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. Archived building spread type 3 was mapped to current University of California, Berkeley/University of Maryland building-spread formulation (type 2). The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

| Choice | Evidence class | Basis and limitation |
|---|---|---|
| Landscape, weather, ignition, Monte Carlo, spotting, and WUI values | documented | Retained from the received archive; upstream derivation and event-specific tuning history are unknown. |
| Model-setting compatibility | derived | Required by the choices supported by the inspected model formulation; no physical values were tuned. |
| Comparison time | documented | Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent. |
| $p \geq 0.50$ footprint | selected for this comparison | Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use. |
| VIIRS hull ratio 0.2 | selected for this comparison | Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter. |

Table 2: Provenance of consequential configuration and comparison choices.

## 5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

## 6 Validation metrics and comparison method

For member $i$, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and $t_c$ the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c] \,, \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support $O$, intersection $I = |P \cap O|$, and valid-cell count $M$, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and $F_1$ balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

## 7 Simulation results and field-level comparison

The current scientific status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

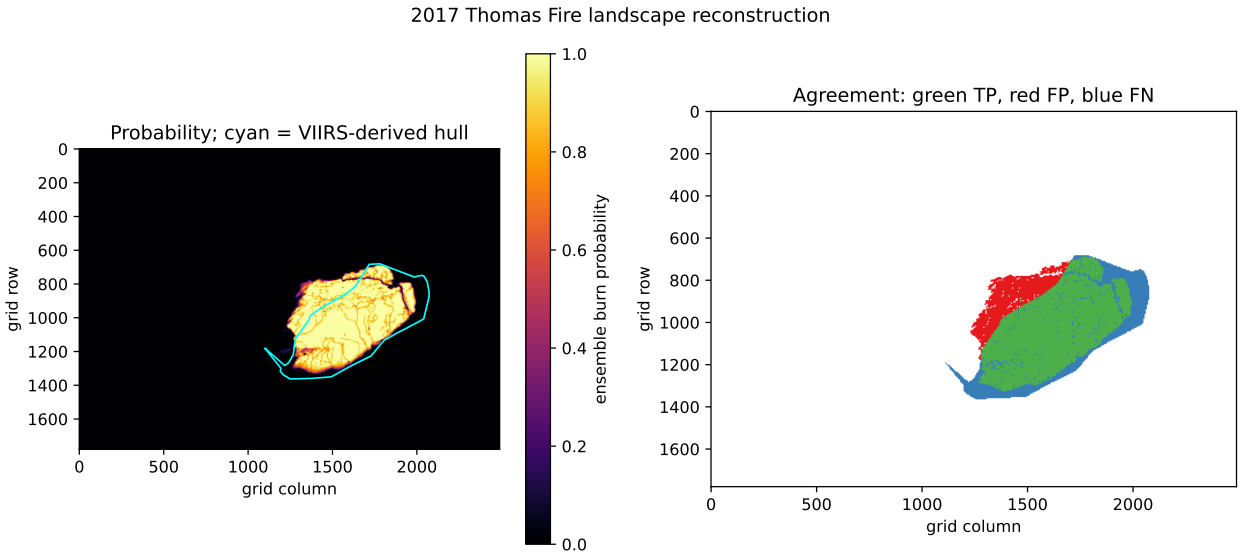



Figure 5: Ensemble burn probability, VIIRS-derived support, and cell-level true-positive, false-positive, and false-negative regions.

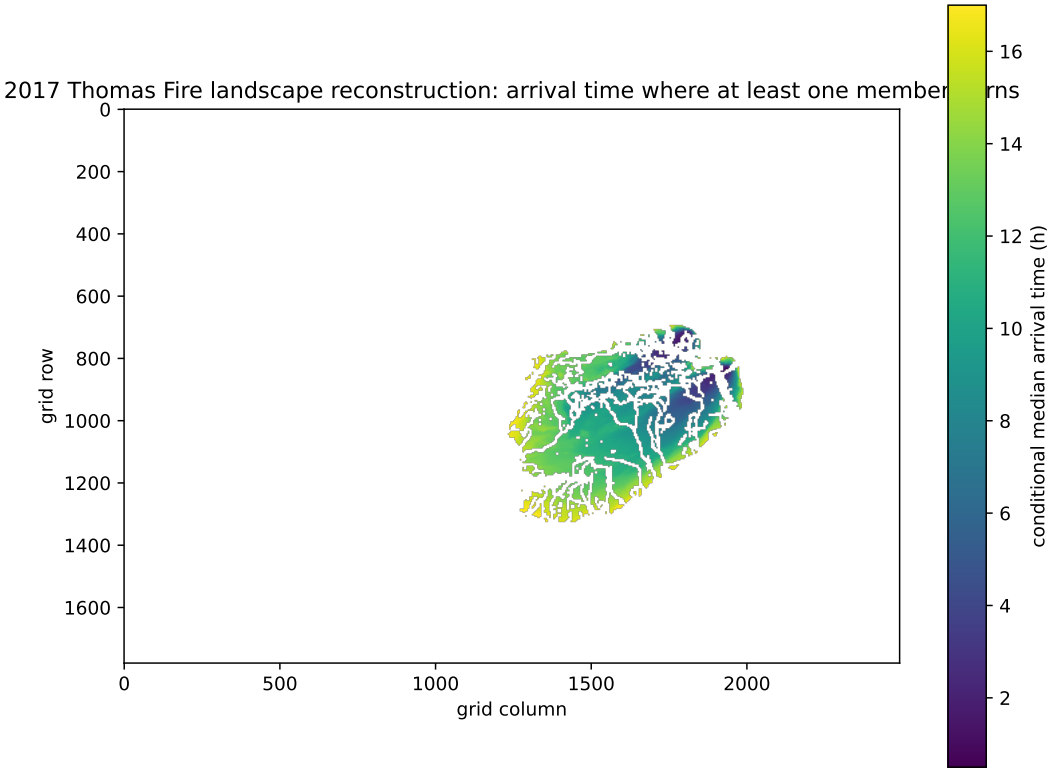



Figure 6: Conditional median arrival time among members that burn each cell by the comparison time. Conditioning is stated because it omits no-burn members.

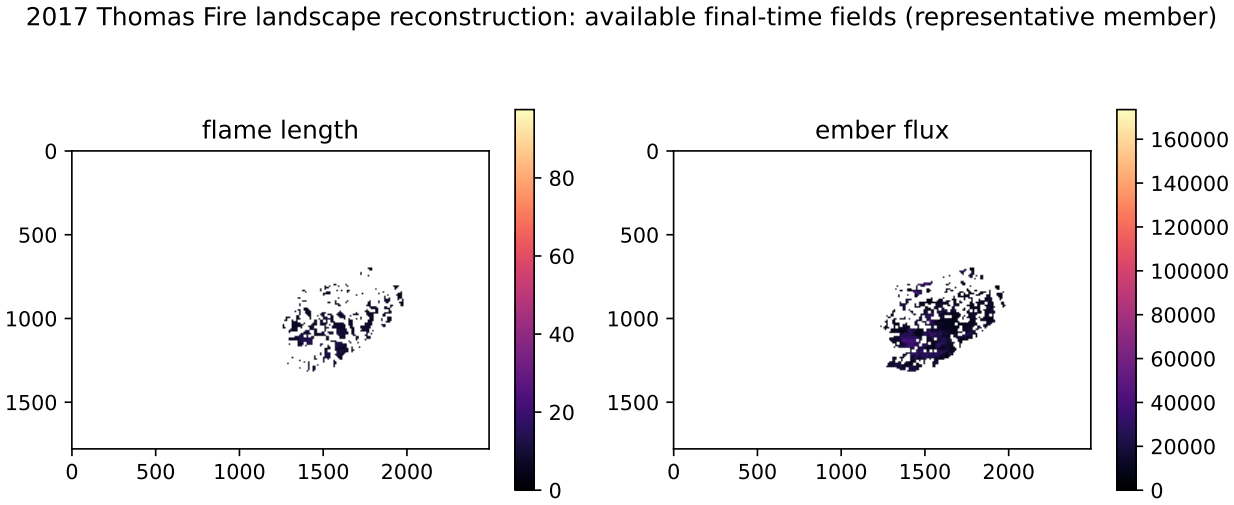



Figure 7: Applicable final-time two-dimensional fields found by the postprocessor. Omission of this panel means no recognized optional field was available; it does not imply a zero field.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported and must not be inferred from WUI input rasters alone.

# 8 Integrated statistics and calculated metrics

| Metric | Current value |
|---|---|
| Jaccard overlap | 0.6214 |
| Precision | 0.7985 |
| Recall | 0.7369 |
| $F_1$ score | 0.7665 |
| Brier score | 0.03336 |
| Predicted/observed area ratio | 0.9227 |
| Predicted area (km$^2$) | 266.8 |
| Observation-support area (km$^2$) | 289.1 |
| Ensemble members found | 50 |

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

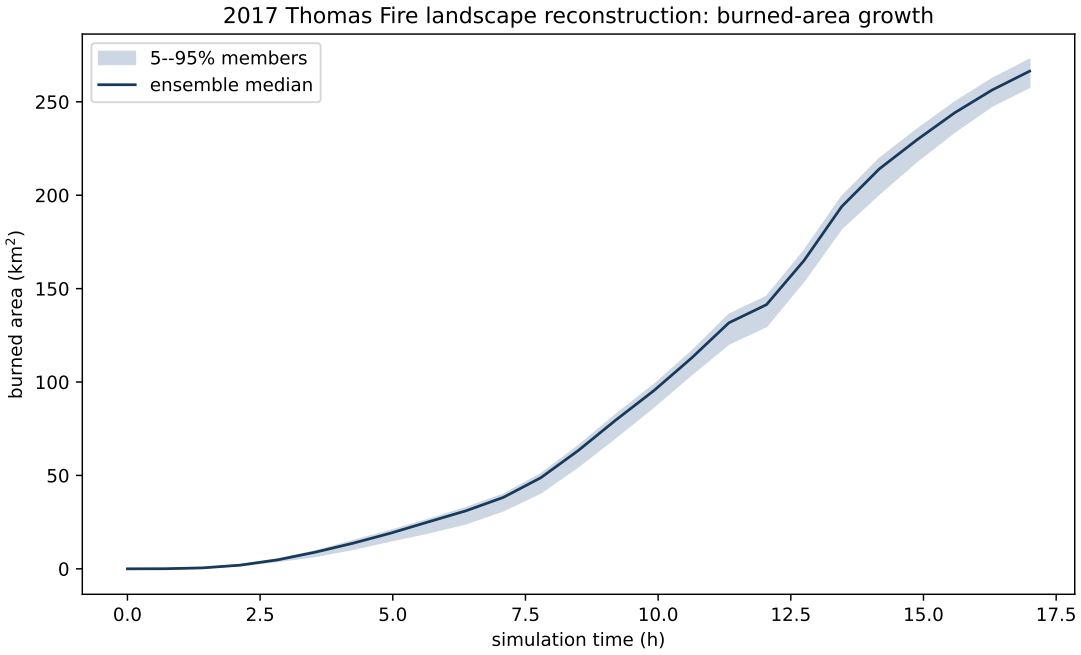



Figure 8: Median and 5th–95th percentile ensemble burned-area growth.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

# 9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish

spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

# 10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **CHARACTERIZED**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.

# 2017 Tubbs Fire Landscape Reconstruction

## 1 Validation objective and scope

This case asks whether ELMFIRE can reconstruct the landscape-scale spatial growth of the 2017 Tubbs Fire Landscape Reconstruction through 12 hours when driven by the supplied landscape, weather, moisture, ignition, spotting, and wildland–urban interface (WUI) inputs. The comparison evidence is a time-filtered set of active-fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS). The case exercises the coupled surface- and crown-fire, level-set, firebrand, ember-ignition, Monte Carlo, and building-spread processes. It does not isolate or verify any single algorithm, and it does not establish predictive skill for events outside the three received reconstructions.

The scientific question is deliberately two-sided: does the ensemble overlap the observation-derived footprint, and does it do so without achieving overlap by excessive fire growth? Overlap, omission, commission, area, and probabilistic error are therefore reported together. The archive provides no independent, predeclared acceptance thresholds, so completed runs are *characterized*; they are not labeled PASS or FAIL.

## 2 Reference event, experiment, and observations

The configured event start, expressed in Pacific daylight time (PDT), is 8 October 2017 at 21:45 PDT. The comparison instant is 9 October 2017 at 09:45 PDT (16:45 UTC), with UTC indicating Coordinated Universal Time. The supplied observation dataset contains 1168 point records in the geographic coordinate system defined by EPSG:4326; 597 occur on or before the comparison cutoff. The observation attributes identify the instrument as VIIRS and are consistent with the National Aeronautics and Space Administration (NASA) Fire Information for Resource Management System (FIRMS) active-fire product. NASA describes VIIRS active-fire data as nominal 375-m thermal detections, not mapped fire perimeters. General product information is available from NASA FIRMS active-fire product documentation and NASA Earthdata VIIRS near-real-time product documentation.

The exact FIRMS collection, granules, download URL, acquisition date, filtering history, and license are not recorded in the case material. Consequently the NASA attribution is inferred from the observation attributes rather than established by complete provenance. Integrity checks identify the supplied observations, which remain unchanged during preprocessing.

For raster comparison, detections through the cutoff are projected to the simulation coordinate reference system (CRS) and enclosed by a concave hull with ratio 0.2 and no holes. If the installed geometry library lacks that operation, the postprocessor reports and uses a convex-hull fallback. This converts intermittent points into a comparison support; it is a declared comparison assumption, not an observed final perimeter. Cloud, smoke, overpass timing, geolocation error, false detections, and the hull choice remain observation uncertainties.

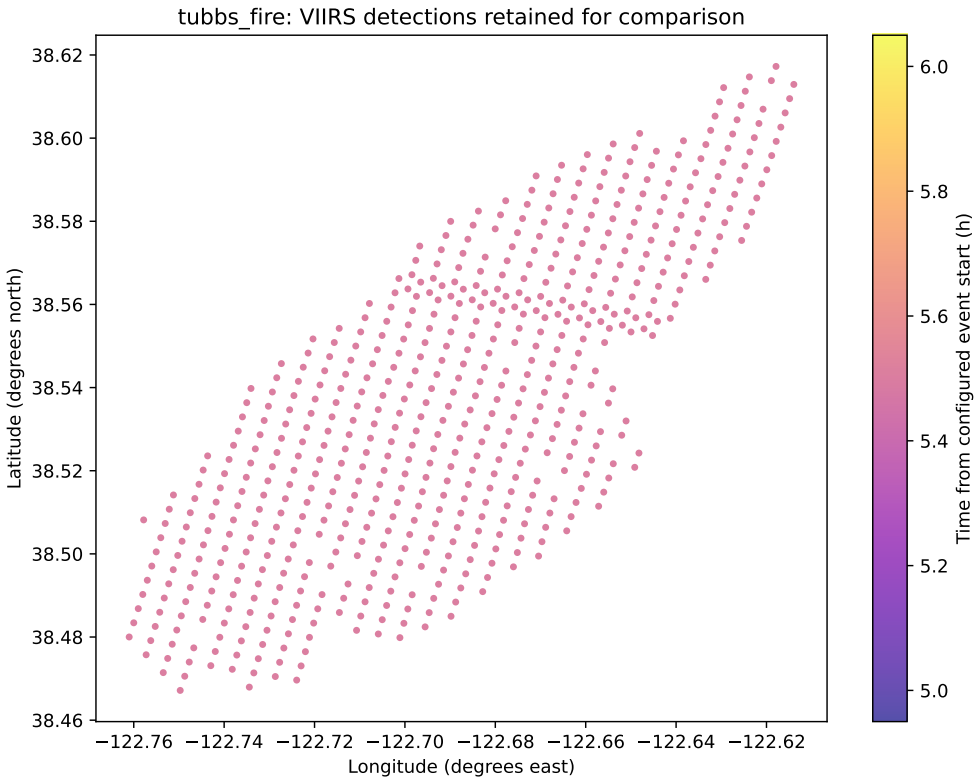



Figure 1: Supplied VIIRS detections retained through the comparison cutoff, colored by elapsed time from the configured event start.

## 3 Input data provenance and characterization

The supplied historical-fire reconstruction is identified by the SHA-256 checksum `e97bf430d251220 19211405f62f98979e929f80bc3f20d8c3b9f1f9f1a8a4955`. The extracted source data are retained unchanged, and their identities are checked before analysis. The collection's only supplied provenance note is "From Dwi." Upstream product names, providers, vintages, acquisition dates, processing history, and licenses for fuels, terrain, meteorology, moisture, WUI, and ignition data are unknown. The report therefore describes the archived values but does not infer unsupported product identities such as LANDFIRE or a specific weather analysis.

| Field | Minimum | Mean | Standard deviation | Maximum |
|---|---|---|---|---|
| Elevation | 15 | 164.8 | 130.8 | 722 |
| Slope | 0 | 8.329 | 8.383 | 47 |
| Wind speed | 0.0608 | 24.32 | 13.19 | 54.81 |
| 1-h moisture | 4.225 | 5.335 | 0.3679 | 6.793 |
| 10-h moisture | 5.225 | 6.335 | 0.3679 | 7.793 |
| 100-h moisture | 6.225 | 7.335 | 0.3679 | 8.793 |

Table 1: Input statistics over valid cells and the 25 weather bands selected by the namelist. Wind speed is interpreted as 20-ft mph and dead-fuel moisture as percent from the archived flags and inspected current source; upstream metadata that would independently confirm these units is absent.

The 300-m weather rasters cover the same extent and CRS as the 30-m landscape but intentionally do not share its pixel grid. The preprocessor checks weather extent, band count, and CRS separately rather than treating the resolution difference as an error.

Aspect and wind direction are circular quantities. Their arithmetic means are not used as directional summaries; the circular means are 351.5° and 40.21°, respectively. Exact means and standard deviations use every valid selected value. Reported quantiles use a deterministic nearest-neighbor sample of at most $512 \times 512$ cells per band, and are labeled accordingly.

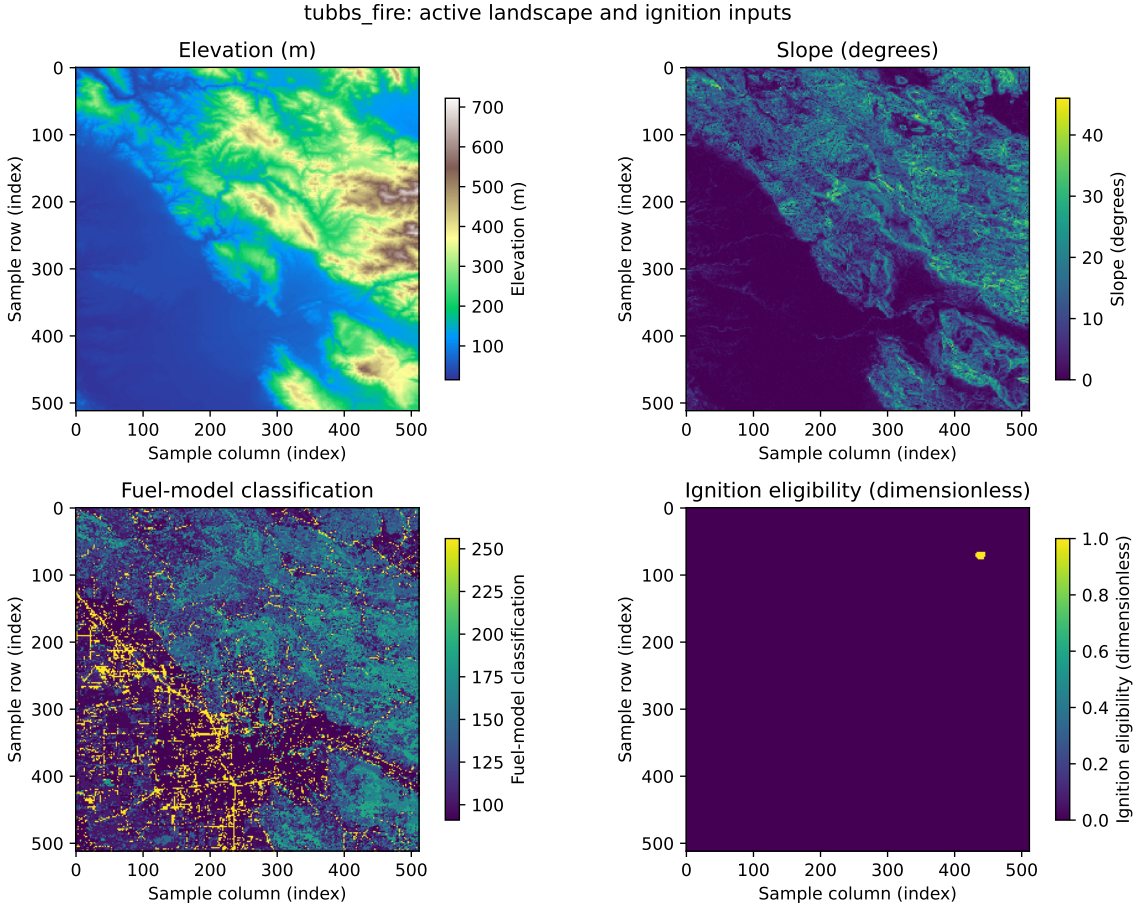



Figure 2: Active elevation, slope, fuel-model, and ignition-mask inputs. These plots characterize model forcing; they are not validation evidence.

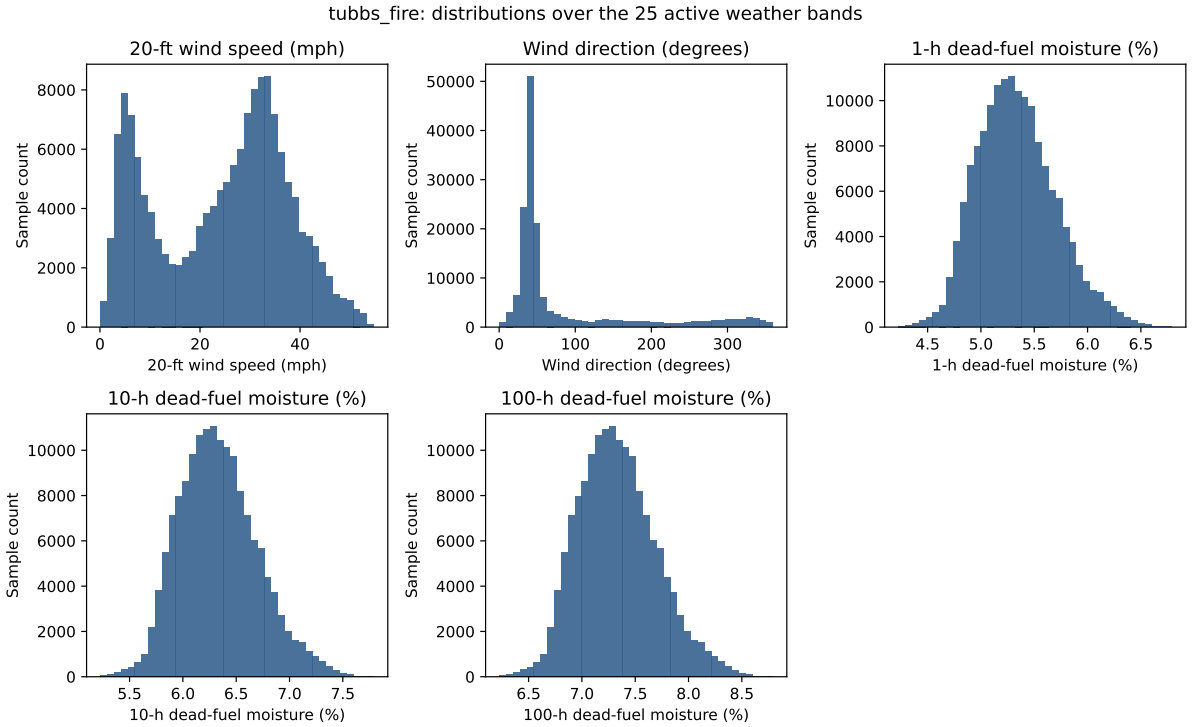



Figure 3: Distributions sampled deterministically from the 25 weather and dead-fuel-moisture bands selected by the namelist.

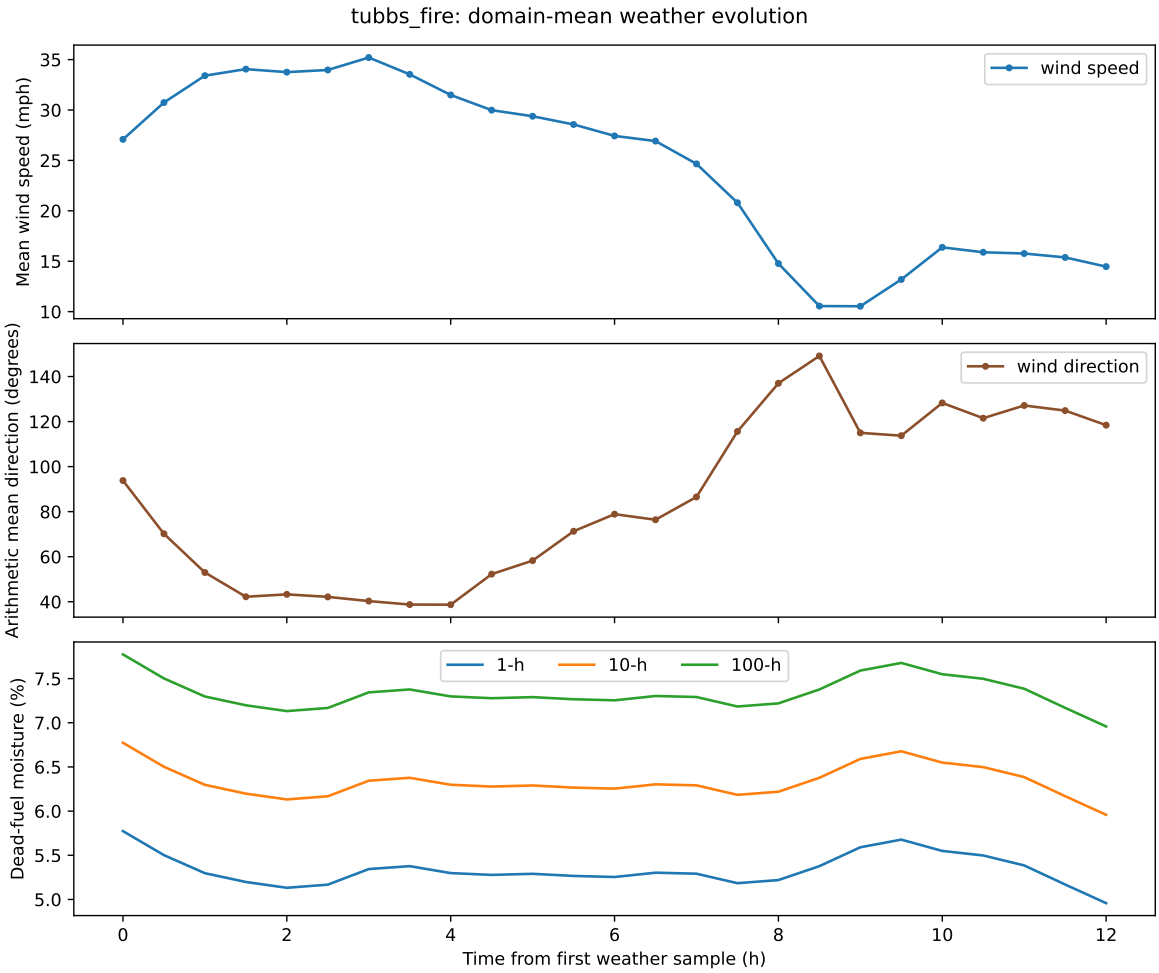



Figure 4: Exact domain-mean wind and dead-fuel-moisture values by active band.

## 4 Model configuration and rationale

The active domain is 700 by 900 cells at 30 m in the archived Lambert conformal conic CRS. Weather is 25-band, 300 m weather rasters at 30-minute intervals, and the ignition treatment is random ignition sampling from the archived ignition mask, with seed 2024. The simulation uses 50 members, a target Courant–Friedrichs–Lewy (CFL) number of 0.45, crown-fire model 2, and fire-power-proportional firebrand generation with empirical transport distance, Eulerian accumulation, and physical ignition. These physical values, seeds, uncertainty ranges, ignition choices, and WUI settings are retained from the received archive except for the documented removal of a contradictory fixed Tubbs ignition. In the available material they are *documented by configuration* but their scientific selection rationale is unsupported; they must not be described as independently derived or literature-justified.

The original settings were adapted only where required for compatibility with the inspected ELMFIRE formulation at source revision `efa0b8fa3935ea4cd30a2bd054c7e9b83db81163`, which already contained additional modifications when inspected. Obsolete domain-description and diagnostic settings were omitted, and the heat-release-rate ellipse adjustment was associated with the wildland–urban interface settings. The earlier spotting choices were expressed as fire-power-proportional generation, empirical transport distance, Eulerian accumulation, and physical ignition. Archived building spread type 3 was mapped to current University of California, Berkeley/University of Maryland building-

spread formulation (type 2). The contradictory archived fixed ignition was disabled because current ELMFIRE applies it in addition to the configured random ignition-mask sample, whereas this case specifies the latter alone. The required physical values are retained in the case specification; this compatibility review does not establish their scientific justification.

| Choice | Evidence class | Basis and limitation |
|---|---|---|
| Landscape, weather, ignition, Monte Carlo, spotting, and WUI values | documented | Retained from the received archive; upstream derivation and event-specific tuning history are unknown. |
| Model-setting compatibility | derived | Required by the choices supported by the inspected model formulation; no physical values were tuned. |
| Comparison time | documented | Fixed in the case specification and consistent with the supplied reconstruction procedure; original scientific rationale is absent. |
| $p \geq 0.50$ footprint | selected for this comparison | Majority-ensemble interpretation used for reproducibility; it is not an acceptance threshold and sensitivity should be reported for scientific use. |
| VIIRS hull ratio 0.2 | selected for this comparison | Compact cumulative support for point detections; not calibrated to an uncertainty-bounded perimeter. |

Table 2: Provenance of consequential configuration and comparison choices.

## 5 Preprocessing and reproducibility

Preprocessing verifies the supplied reconstruction and observations by size and SHA-256 checksum. It checks that required landscape rasters share a grid, confirms that the namelist-selected weather rasters contain at least 25 bands, computes input and observation statistics, and regenerates the four input figures. The resulting statistics and figures characterize the source data without modifying the original evidence.

Simulation is followed by analysis of the complete ensemble. The declared default allocation is 50 Message Passing Interface (MPI) processes, one per ensemble member, without an additional coordinating process. The original simulation results remain available for examination. Reproducible input identity does not by itself guarantee result reproducibility: each run must also record the actual ELMFIRE revision, compiler, dependencies, and platform.

## 6 Validation metrics and comparison method

For member $i$, let $\tau_i(\mathbf{x})$ be the time-of-arrival raster and $t_c$ the declared comparison time. The ensemble burn probability and thresholded footprint are

$$p(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[0 < \tau_i(\mathbf{x}) \leq t_c], \qquad P = \{\mathbf{x} : p(\mathbf{x}) \geq 0.50\}. \tag{1}$$

For the rasterized observation support $O$, intersection $I = |P \cap O|$, and valid-cell count $M$, the analysis reports

$$J = \frac{|P \cap O|}{|P \cup O|}, \qquad \text{precision} = \frac{I}{|P|}, \qquad \text{recall} = \frac{I}{|O|}, \tag{2}$$

$$F_1 = \frac{2\,\text{precision}\,\text{recall}}{\text{precision} + \text{recall}}, \qquad R_A = \frac{|P|}{|O|}, \qquad \text{Brier} = \frac{1}{M}\sum_{\mathbf{x}}(p - y)^2. \tag{3}$$

Jaccard is a symmetric footprint-overlap summary. Precision diagnoses commission/overgrowth, recall diagnoses omission/undergrowth, and $F_1$ balances those errors. The area ratio exposes size compensation that overlap alone can hide. The Brier score retains ensemble probabilities, although the large unburned domain can dominate it. Burned-area growth and conditional median arrival time add temporal context. VIIRS detections and their derived hull are the observational baseline; no previous simulation result is used as truth.

Metrics are calculated only when exactly 50 uniquely identified members reach the comparison time and share one grid and CRS. This completeness requirement prevents partial runs from being reported as scientific results. The static aspect raster defines the analysis-domain mask. Member-specific NoData cells in time-of-arrival outputs represent cells not burned by that member; they are therefore retained as zero burn outcomes rather than treated as a grid-mask mismatch.

## 7 Simulation results and field-level comparison

The current scientific status is **CHARACTERIZED**. Metrics are descriptive because no independent predeclared acceptance thresholds are available. Metrics and figures are generated only from the complete, current ensemble. If that ensemble is unavailable, the analysis reports the result as not evaluable rather than substituting legacy outputs or zero-valued metrics.

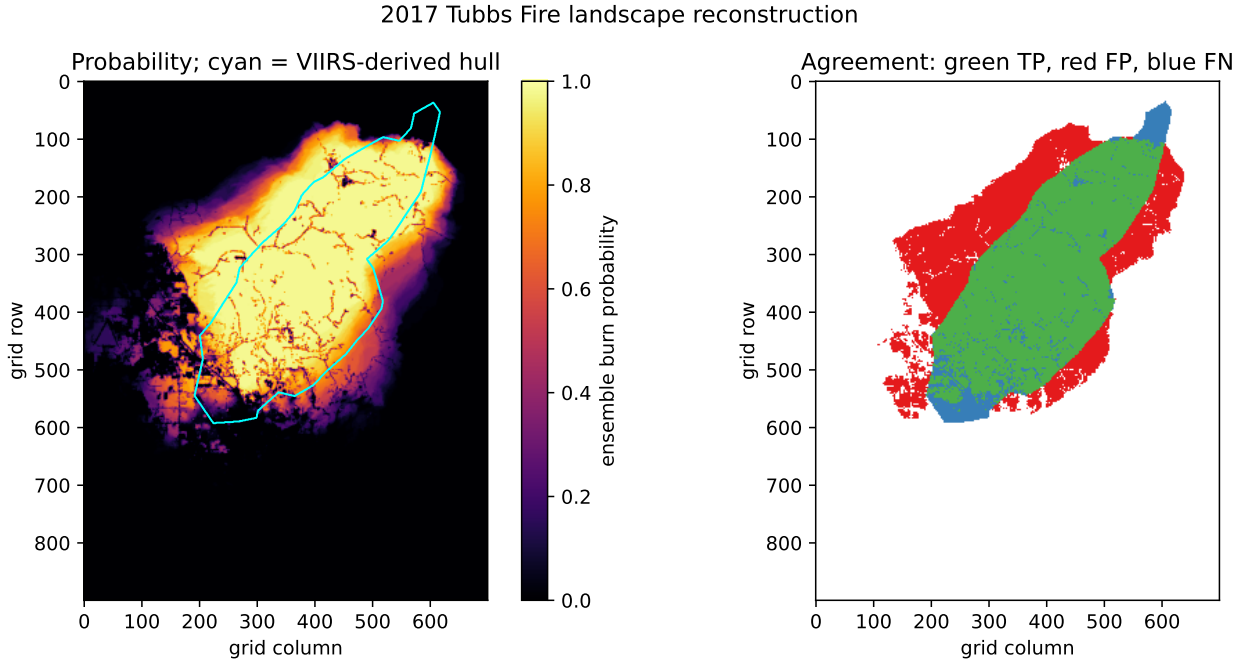



Figure 5: Ensemble burn probability, VIIRS-derived support, and cell-level true-positive, false-positive, and false-negative regions.

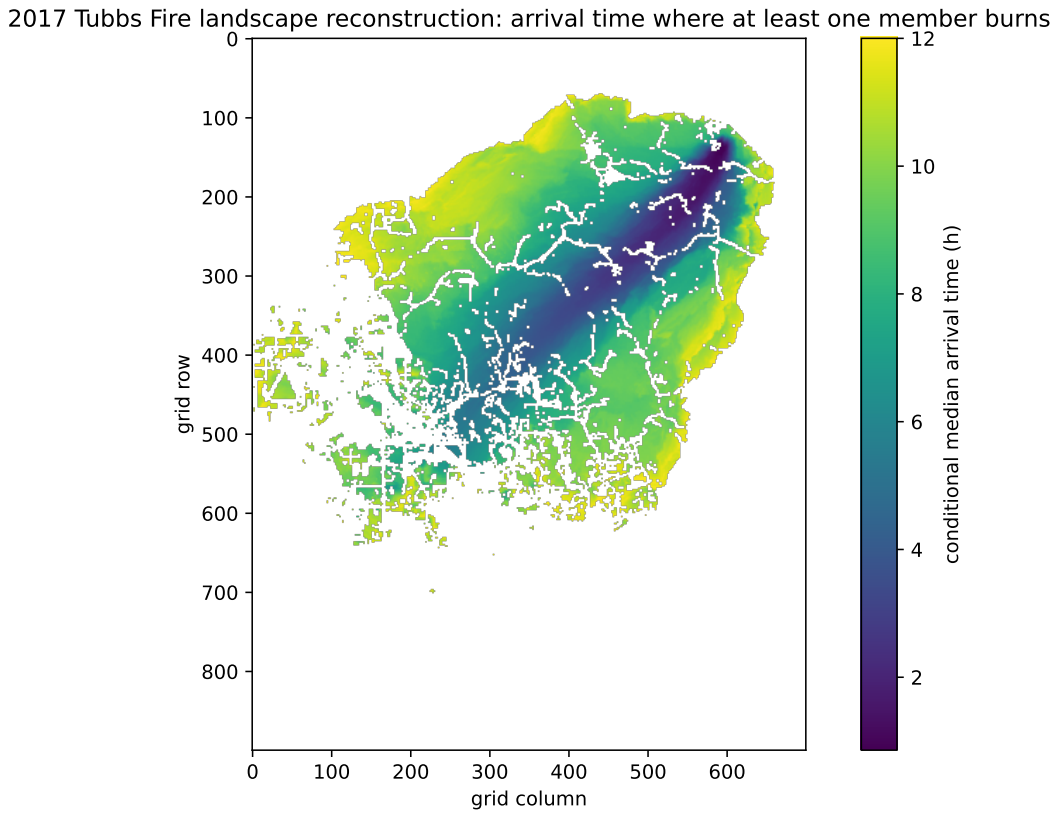



Figure 6: Conditional median arrival time among members that burn each cell by the comparison time. Conditioning is stated because it omits no-burn members.

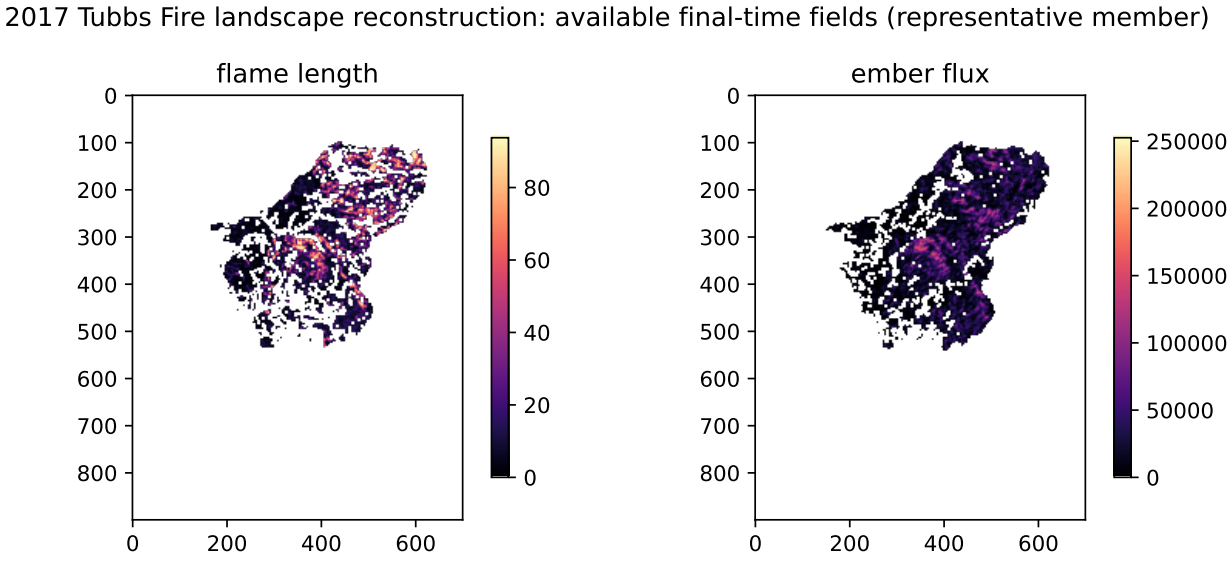



Figure 7: Applicable final-time two-dimensional fields found by the postprocessor. Omission of this panel means no recognized optional field was available; it does not imply a zero field.

The simulation requests flame length and ember flux in addition to arrival time. It does not request a structure-damage count or damage-time output, so a damaged-structures time series is not reported

and must not be inferred from WUI input rasters alone.

## 8 Integrated statistics and calculated metrics

| Metric | Current value |
|---|---|
| Jaccard overlap | 0.555 |
| Precision | 0.6051 |
| Recall | 0.8702 |
| $F_1$ score | 0.7138 |
| Brier score | 0.09008 |
| Predicted/observed area ratio | 1.438 |
| Predicted area (km$^2$) | 133.8 |
| Observation-support area (km$^2$) | 93.02 |
| Ensemble members found | 50 |

Table 3: Calculated statistics. **MISSING** is intentionally conspicuous when the completeness requirement prevents evaluation.

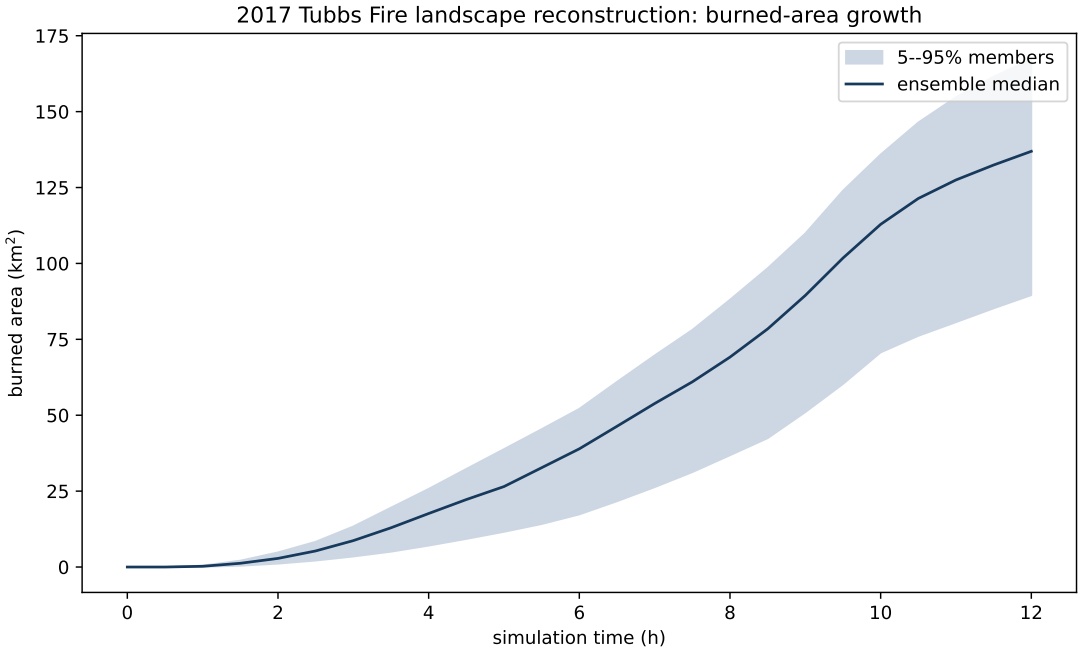



Figure 8: Median and 5th–95th percentile ensemble burned-area growth.

These values are compared with the observation-derived support and not with a stored ELMFIRE baseline. Because neither metric thresholds nor observation uncertainty bounds were declared independently, a populated table supports scientific characterization only.

# 9 Interpretation and uncertainty

A low Jaccard value with high recall and low precision indicates spatial overgrowth; high precision with low recall indicates undergrowth or a displaced trajectory. An area ratio near one cannot establish spatial agreement. The Brier score must be read with the footprint metrics because correct prediction of the large unburned domain may obscure errors near the fire.

The dominant limitations are incomplete input provenance, an observation product that samples thermal anomalies rather than perimeter, uncertain hull sensitivity, unknown event-specific calibration history, and absence of predeclared acceptance bounds. Model-form error, uncertain fuels and moisture, meteorological representation, ignition timing/location, spotting parameters, and WUI parameters are not separable using these aggregate metrics alone. The 50 archived perturbations characterize one configured ensemble; they do not by themselves quantify all epistemic uncertainty.

# 10 Conclusions and limitations

This case characterizes the agreement between a complete simulated ensemble and a spatial support derived from active-fire detections. Input statistics describe the forcing, spatial comparisons distinguish omission and commission, and integrated quantities provide context for the overlap metrics. Its current scientific result is **CHARACTERIZED**; the accompanying reason states whether the complete current ensemble supported quantitative characterization.

To support a defensible validation decision, future case owners must supply the upstream input citations and licenses, exact VIIRS collection/granule and acquisition provenance, the derivation or literature basis for ignition, spotting, WUI, and perturbation settings, disclosure of any calibration against this event, and acceptance thresholds fixed before inspecting the evaluated run. Until then, even a completed run is correctly labeled CHARACTERIZED, not PASS or FAIL.